Copernicus Marine Service

QUALITY INFORMATION DOCUMENT

North West European Shelf Production Centre
NWSHELF_ANLYSIS_FORECAST_WAV_004_014

Issue: 4.0

Contributors: Christopher Stokes, Andy Saulter, Richard Davies, Jonathan Tinker

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CHANGE RECORD

When the quality of the products changes, the Quality Information Document (QuID) is updated, and a row is added to this table.

The third column specifies which sections or sub-sections have been updated. The fourth column should mention the reason for updating the document. Column 5 should include the names of the author(s) doing the changes. Including the name in column 6 shows that the document has been validated by the TAC/MFC PQ leader or deputy (or TAC/MFC PQ leader or the deputy). This validation means the document was checked for contents, formats and consistency with other documents.

Issue	Date	§	Description of Change	Author	Validated By
1.0	12/09/2018		Preliminary Draft	Andy Saulter	Marina Tonani
1.0	30/11/2018		Release Document	Andy Saulter	CMEMS PQ Coordination Team
1.1	28/08/2020		Science Update – Coupled Ocean-Wave Model	Andy Saulter	Marina Tonani, Nieves Valiente
2.0	31/07/2023		**[EIS September 2023]** Complete system update implying change of the modelling system configuration and resolution.	Stefania Ciliberti, Cristina Toledano, Lotfi Aouf, Alice Dalphinet, Roland Aznar	Sylvain Cailleau and Marcos Garcia Sotillo
3.0	30/08/2024		**[EIS November 2024]** New product at 1/36° resolution	B. Gomez, L. Aouf, A. Dalphinet, A. Alonso, S. Ciliberti	S. Cailleau, M. G. Sotillo
4.0	22/05/2025		**[EIS November 2025]** Re-entry into service of Met Office coupled wave-physics analysis forecast system	Christopher Stokes, Andy Saulter, Richard Davies, Jonathan Tinker	Christine Pequignet and Thorger Brüning

ACRONYM TABLE

Acronym	Description
AMM15	Atlantic Margin Model, 1.5 km resolution
EAN	Estimated Accuracy Number
IBI	Iberia-Biscay-Ireland
MFC	Copernicus Marine Service Monitoring and Forecasting Centre
MY	Multi-year
NRT	Near Real Time
NWS	North West Shelf
TAC	Copernicus Marine Service Thematic Assembly Centre
WAV	Wave product
WWIII	Wavewatch 3 wave model

KEY DEFINITIONS

Analysis	Incorporation of an observation data assimilation step to a model prediction system in order to achieve an optimal estimate of the true state of the atmospheric or marine system at a given point in time.
Average zero-upcrossing period (model VTM02)	Average time interval between consecutive waves as they cross the mean sea level in an upward direction.
Average wave direction (model VMDR)	Average direction of wave approach in a given sea state.
Coastal Waters	Approximately 5 km or less from the coastline and where bathymetric variability is less than a kilometre in scale.
Forecast	A simulation that uses a numerical model to predict a future period (typically days-weeks in the future)
Hindcast	A simulation that uses a numerical model to recreate a historical period (typically days-decades in the past)
Match-up	Co-location in space and time between a model prediction and a real-world measurement.
Open Waters	Approximately 5 km or more from the coastline and where bathymetric variability is on the scale of a kilometre or greater.
Peak wave period (model VTPK)	Interval in seconds between wave crests associated with the peak wave energy frequency.
Significant wave height (model VHM0)	Average height of the largest 1/3rd of waves in a given sea state. Computed using sea surface elevations (four times the standard deviation of the surface elevation) or from the zeroth-order moment of the wave spectrum from a wave model.

RELEVANT DOCUMENT LIST

Ref	Title	Version
CMEMS-NWS-PUM-004-014	Product User Manual for the corresponding product NWSHELF_ANALYSISFORECAST_WAV_004_014.	3.0
CMEMS-NWS-PUM-004-013	Product User Manual for equivalent NWS physical product NWSHELF_ANALYSISFORECAST_PHY_004_013 which uses the same bathymetry and domain as the WWIII-AMM15 configuration described in this document.	3.0
CMEMS-NWS-QUID-004-013	Quality Information Document for the equivalent NWS physical product NWSHELF_ANALYSISFORECAST_PHY_004_013 which uses the same bathymetry and domain as the WWIII-AMM15 configuration described in this document.	3.0

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I EXECUTIVE SUMMARY

I.1 Products covered by this document

This document covers the NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004-014 product, which addresses the ocean surface wave characteristics from the North-West European Shelf analysis and the forecast modelling system at approximately 1.5 km resolution.

It should be noted that between September 2023 and November 2025, the North West Shelf-wave products (NWS-WAV) were delivered by the IBI-MFC. From the November 2025 release, the NWS-WAV products are provided by the NWS-MFC using their North West Shelf model applications to generate NWS NRT products for the Copernicus Marine Service.

Also, whilst the North-West Shelf wave product is provided on a regular grid resolved at approximately 1.5 km, it is derived from a model with a native grid using a variable resolution (3 and 1.5 km cells) and formulation on a rotated pole. The wave model is run in this form for purposes of computational efficiency and to ensure product timeliness. The 1.5 km resolution of the model coastal cells restricts the applicability of the information provided in this document to open waters approximately 5 km or more from the coastline and where bathymetric variability is on the scale of a kilometre or greater.

I.2 Summary of the results

I.2.1 Hindcast model quality

The quality of the hindcast component of the North-West Shelf MFC wave product NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004_14 has been assessed via a comparison of the underpinning wave model (WAVEWATCH III covering the AMM15 domain) against observations over the period 01/01/2022 to 31/12/2023. The assessment of the hindcast is the best way to understand the model quality, since the wave analysis-forecast system provided to the CMEMS will run without data assimilation[1].

The evaluation in this QuID identifies both ***'open waters'*** (offshore) and ***'coastal'*** (nearshore) observations data sets and presents the model quality in these waters separately. The quality assessment covers a subset of key wave parameters, results of which are summarised below:

Significant wave height: The assessment of significant wave height tests the wave model's ability to represent the bulk energy imparted to ocean surface waves from the atmosphere. In open waters, the model shows good agreement with both in-situ and satellite altimeter observations; the model and observations are correlated at a minimum level of r = 0.96, or even higher, and shows relatively low bias overall (on the order of 0.1 m). Regionally, the standard deviations of differences between the model and observations in open waters fall within 10-17% of the observed mean significant wave height, and 18-26% of the observed standard deviation. In the majority of coastal waters, the model and observations are correlated in the range r = 0.85-0.98, with low overall bias (regionally 0.03, up to 0.10 m in the Thames Approaches). Standard deviations of differences between the model and observations are approximately 14-29% of the mean observation and 19-69% of the observed standard deviation. The largest wave height biases and/or standard deviations of differences occur within estuary approaches (for example, the Thames Approaches and Bristol Channel in the UK), where relatively small waves are influenced by large tidal ranges and forcing winds may have decreased accuracy due to local coastal effects.

[1] a wave hindcast forced by short-range forecast winds and analysis surface currents is used to constrain initial conditions

Average zero-upcrossing period: These tests assess the wave model's ability to represent the overall distribution of wave energy through the frequency domain of the wave spectrum, with an emphasis on higher frequencies due to the use of the model spectrum's second frequency moment in calculation of this parameter. A quality assessment of average zero-upcrossing period could only be carried out for UK coastal locations, due to a lack of available observations further offshore. In coastal waters, correlation is in the range r = 0.57-0.91, with overall bias of -0.06 s (ranging regionally from -0.41 s to 0.10 s), and standard deviation of the model-observation differences equivalent to approximately 7-20% of the observed mean and 29-98% of the observations' standard deviation. The model generally under-predicts observations below 5 seconds and overpredicts higher wave periods by around 10-15%. The largest wave period biases and/or standard deviations of differences occur within estuary approaches (for example, the Thames Approaches and Bristol Channel in the UK), where relatively small waves are influenced by large tidal ranges and forcing winds may have decreased accuracy due to local coastal effects.

Peak wave period: The quality assessment of peak wave period tests the model's ability to represent wave growth and dissipation processes in the spectral frequency domain, since the peak frequency will shift from higher to lower frequencies as waves grow and the sea-state matures. In open waters few observations of peak wave period were available, but quantiles throughout the peak wave period range were well represented by the model, albeit with poor model-observation correlation (r < 0.52) and a relatively high bias in some regions (up to 3 s) but low bias overall (-0.1 s). In the majority of coastal waters, the model and observations are correlated in the range r = 0.71-0.87, with overall bias of 0.58 s (ranging regionally from -0.10 s to 0.84 s). Standard deviations of model-observation differences are equivalent to approximately 17-46% of the observed mean and 55-122% of the observed standard deviation. Part of the reason for the additional level of error in this parameter lies in its variability in multi-modal sea-states (where two or more spectral peaks may be present), leading to high penalties for small differences between the modelled and observed wave spectrum. This issue is particularly clear in coastal waters with intermittent exposure to Atlantic swell (southwest of the UK and English Channel), where swell events are over-represented. Nevertheless, the overall assessment of the peak period suggests that during energetic events where a sea is dominated by one particular wave component, the peak in the wave spectrum is reasonably well predicted by the model.

Average wave direction: The assessment of this parameter tests the model's ability to represent the distribution of wave energy in directional space. An assessment was made for coastal waters only due to a lack of directional observations further offshore. The model bias is overall < 1 degrees (well below the 10 degrees directional spectral bin width for the model) but ranges regionally (-5.4 to 3.5 degrees); the RMSD is within the range of 17-33 degrees, whilst standard deviation of model-observation differences are equivalent to approximately 6-33% of the observed mean and 28-53% of the observations' standard deviation.

Performance in dynamic coastal regions: Within the coastal quality assessment, one region identified with significantly poorer performance is the Bristol Channel. This is a highly dynamic region, with an extreme tidal range and current speeds and highly variable local fetches. Whilst the model showed some skill in this region for certain parameters/physical conditions, it is recommended that data from this region and similar waters on the North-West Shelf (e.g., waters around the Channel Islands and Gulf of St Malo) are used with some caution.

I.3 Estimated Accuracy Numbers

Table 1 presents a summary of estimates for the mean (model minus observation bias) and standard deviations of errors describing the differences between the model estimates of wave characteristics (derived from the model wave spectrum) and observations (characteristic derivation method is variable)

associated with the model hindcast/analysis. These metrics, along with an assumption regarding the probability distribution function for model-observation errors, can be used to estimate the likelihood of the errors falling within a given range. For example, assuming the errors to be normally distributed, approximately 90% of hindcast significant wave height values can be expected to fall within ±0.5 m of the observed value, and 90% of average zero-upcrossing period model minus observation differences can be expected to fall in the range -1.5 to 0.5 seconds.

Unless stated otherwise, the accuracy estimates are given for the full model domain. Where available, regional breakdowns are provided in Section IV.

Table 1. Baseline model quality for validated parameters

Estimated Variable	Location	Supporting Observations	Number of Observations	Mean difference	Difference Std
Significant wave height* (model VHM0)	Open Waters	In-situ platforms / Satellite altimeters	155,192	0.06 m	0.33 m
Significant wave height (model VHM0)	Coastal	In-situ platforms	747,877	0.03 m	0.23 m
Peak wave period (model VTPK)	Open Waters	In-situ platforms	8,811	-0.12 s	3.81 s
Peak wave period (model VTPK)	Coastal	In-situ platforms	687,824	0.58 s	2.14 s
Average zero-upcrossing period (model VTM02)	Coastal	In-situ platforms	741,498	-0.06 s	0.70 s
Average wave direction (model VMDR)	Coastal	In-situ platforms	687,870	-0.65°	23.01°

* Significant wave height accuracy estimates in open waters use the worst case taken from in-situ and satellite altimeter-based verification. For further details see Section IV.

II PRODUCTION SYSTEM DESCRIPTION

II.1 Production centre's name

The production centre is the North-West Shelf MFC - Met Office, UK.

II.2 Production system name

The production system is the North-West European Shelf Wave Analysis and Forecast (NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004_014).

II.3 System Description

This document details the quality of products from the North-West European Shelf Wave Analysis and Forecast system. These products are provided on a 1.5 km grid, derived from native fields generated by a WAVEWATCH III (version 7.12) at 3-1.5 km Atlantic Margin Model (hereafter denoted as WWIII-AMM15) and at an hourly frequency. The wave model provides a description of ocean surface gravity wave (periods 1.5-25 seconds) characteristics as an extension to the existing physical and ecosystem model products provided by the North-West Shelf MFC. The model became operational within the CMEMS in November 2018 and has been subject to an update in December 2020, in which the model boundary conditions (currents) from the physical ocean model were directly coupled to the wave model rather than read in from files generated by a separate physical model. An alternative wave model product was provided by the IBI-MFC between September 2023 and November 2025. From November 2025, the WWIII-AMM15 entered back into service. The following subsections describe the model component and its dependencies, in terms of models providing forcing and lateral boundary conditions.

II.3.1 Region, grid and bathymetry

The WWIII-AMM15 model is a nested regional wave model configuration. The model's products are defined on an identical grid to the North-West Shelf MFC modelling system for the physical ocean. For computational efficiency and product timeliness, the model itself is formulated using both a rotated North pole (sited at 37.5°N, 177.5°E) and a Spherical Multiple-Cell grid (SMC; Li, 2011; 2012). In this instance the SMC grid uses two tiers of resolution, comprising 3 km cells away from the coastline in water depths greater than 40 m and 1.5 km cells otherwise (e.g., Figure 1, panel a). This grid is derived directly from the 1.5 km Atlantic Margin Model grid used for the physical system (Graham et al., 2018), in order that coastal cells are collocated in both wave and physical systems.

The model is located on the North-West European continental shelf (NWS), covering approximately 45°N, 20°W to 63°N, 13°E. The domain extends beyond the continental shelf in order to place the model's boundary region in the deep waters of the adjacent North-East Atlantic, but the focus region for the model comprises waters of the shelf seas (i.e. (using UK terminology) the North Sea, Irish Sea, English Channel and Celtic Sea). The 1.5 km resolution of the model enables its use in coastal embayments scaled on the order of 10 km or greater, but model utility is more restricted in areas of the coastal zone where gradients in the wave field (for example due to topographic sheltering and strong sub-grid scale variability in shallow water bathymetry and currents) are strong. The AMM15 model and product domains are shown in Figure 2. In this figure, the changes in shading denote regions in the model where wave propagation and dissipation are increasingly affected by shallow water bathymetry. An example significant wave height field from southeast UK waters is shown in Figure 1 (panel b), depicting changes in significant wave height caused by shallow water bathymetry features.

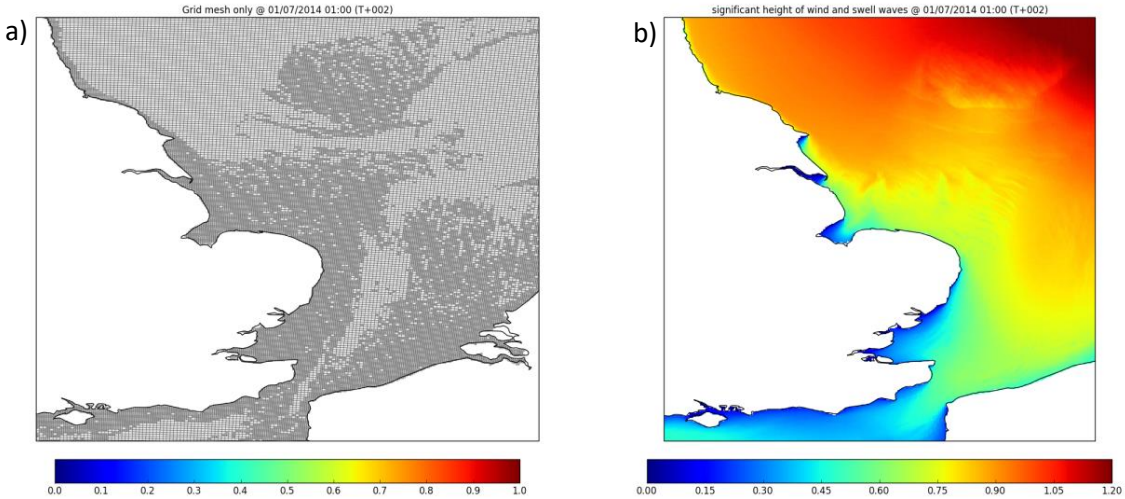

Figure 1: Examples of AMM15 grid mesh showing un-shaded and shaded areas where the mesh is refined from 3 km to 1.5 km, respectively (a), and impact on significant wave height field in southeast UK waters (b).

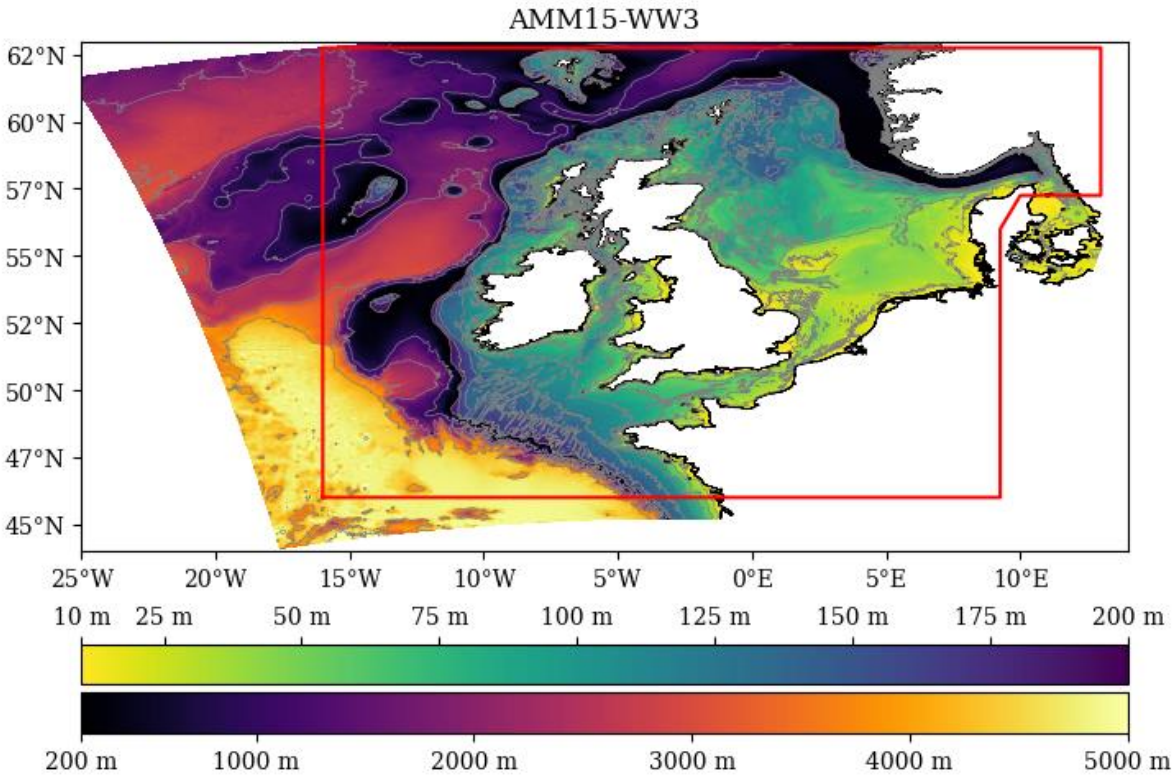

Figure 2: Domain extents (red box) and bathymetric depth (colour scale) for the NORTHWESTSHELF_ANALYSIS_FORECAST_WAV_004-014 product.

Wave models describe ocean surface conditions only, so do not require a vertical coordinate system. Furthermore, the majority of effects on wave propagation and dissipation occur in waters of

intermediate and shallow depth, as defined relative to the wave frequency (inverse of wave period, Equation 1):

$$d \leq \frac{g}{4\pi f^2} \qquad \text{(Eq. 1)}$$

where *d* is water depth, *g* acceleration due to gravity and *f* is wave frequency.

Thus, bathymetry is only a controlling mechanism on the wave field for depths shallower than approximately 490 m, based on a minimum frequency in the model of approximately 0.04 Hz (period 25 seconds). The WWIII-AMM15 configuration uses the same bathymetry as the equivalent NWS physical model (see CMEMS-NWS-QUID-004-013), which has been derived from EMODNet 2015 bathymetry[2] and corrected to mean sea level. In order to match the grid set-up in the hydrodynamic model and improve the estimate of wave energy transmission in the vicinity of sub-grid scale topographic features such as small islands and coastal headlands, a number of sub-grid blocking cells can be defined for the wave model following the method described by Chawla and Tolman (2008).

II.3.2 Spectral grid

WWIII-AMM15 wave spectra are defined on a two-dimensional frequency-direction grid using 36 (10°) direction cells and 30 geometrically spaced frequency cells. The frequency spacing is used in order to enable an efficient application of the model's Discrete Interaction Approximation parameterisation of nonlinear wave-wave interactions (Hasselmann et al., 1985) and is based on a multiplication factor of 1.1. The minimum frequency for the configuration is set at 0.4118 Hz, such that the maximum frequency bin is at 0.6532 Hz and the prognostic part of the wave model covers periods from approximately 25 to 1.5 seconds. In order to include the important contribution of higher frequency waves to wave growth/dissipation processes and the output wave characteristics, a parametric tail is fitted for frequencies above the spectral maximum (e.g., WAMDIG, 1988); in this case, using an f^{-5} relationship (Tolman, 2014).

II.3.3 Wave model and source term physics configuration

WAVEWATCH III (Tolman, 1997, 1999, 2002a, 2009, 2014; The WAVEWATCH III Development Group, 2016) is a community wave model[3] maintained and released by the United States National Center for Environmental Prediction (NCEP). WWIII-AMM15 is based on version 7.12 code (Tolman, 2014), with some modifications made to post processing modules in order to produce wave partition parameters specific to CMEMS.

The model predicts the evolution of the two-dimensional (frequency-direction) wave energy spectrum in time and space. In order to calculate the spectral changes on the moving reference frame created when waves are superimposed on the surface current field, the model solves the action-balance equation (Equation 2):

$$\frac{DN}{Dt} = \frac{S}{\sigma} \qquad \text{(Eq. 2)}$$

[2] See: http://www.emodnet-bathymetry.eu/

[3] See the model and release documentation at http://polar.ncep.noaa.gov/waves/wavewatch/

where *DN* represents the change in wave action-density spectrum, *Dt* is change in time, σ is the radian wave frequency, and *S* represents the sum of terms describing sources and sinks of wave energy.

The left-hand side of the Equation 2 denotes the propagation of wave energy in time and space. Propagation of wave energy in WWIII-AMM15 is solved using the second order upstream non-oscillatory scheme discussed by Li (2008) and incorporating a hybrid 'garden sprinkler effect' alleviation scheme which uses both swell age diffusion and an averaging scheme (WAVEWATCH III Development Group, 2016).

S is broken down into a number of 'source terms' that define wave growth and dissipation. In the case of WWIII-AMM15, four key types of term are identified describing:

1. inputs to wave energy (S_{in}) via momentum stress transferred from the atmosphere;
2. dissipation of wave energy via whitecapping (S_{ds});
3. transfer of energy between wave frequencies via wave-wave interactions (S_{nl}); and
4. dissipation of wave energy due to the interactions between waves and the underlying sea bed (e.g., bottom friction S_{bot}, and depth induced wave breaking S_{db}).

In the WWIII-AMM15, the source term scheme representing input and dissipation of wind waves follows Ardhuin et al. (2010). Nonlinear wave-wave 'quadruplet interactions', which shift wave energy toward lower frequencies, are parameterised using the Discrete Interaction Approximation of Hasselmann et al. (1985). Shallow water effects cannot be fully resolved at the model's 3-1.5 km resolution but are partially parameterised using bottom friction dissipation following the JONSWAP formula (Hasselmann et al., 1973) and depth induced wave breaking following Battjes and Janssen (1978). In general, the tuning constants for these parameterisations follow the default settings noted in Tolman (2014). Table 2 describes those constants that have been adjusted in order to optimise performance of WWIII-AMM15 specific to the North-West Shelf region and for Met Office Unified Model wind forcing.

Table 2: WAVEWATCH III source term tuning parameters revised from version 4.18 defaults (Tolman, 2014)

Namelist	WWIII variable	Default value	WWIII-AMM15 value
ST4	BETAMAX	1.52	1.39
SBT1	GAMMA	-0.067	-0.038
SDB1	BJALFA	1.0	0.2

II.3.4 Forcing and lateral boundary conditions

Forcing for the wave model is taken from two sources. Hourly surface (10 m) wind data are provided from the high-resolution atmosphere-ocean coupled global configuration (Williams et al., 2018) of the Unified Model produced by the Met Office (UM; e.g., Walters et al., 2019; Brown et al., 2012). At the time of the model release, the wind field is defined on an approximately 10 km resolved grid over the AMM15 region. Using the presently available SMC wave model method for applying forcing fields, pre-processed wind fields are interpolated to the level of the coarsest wave model cells, equivalent to approximately 3 km.

Surface current effects on the wave field are included by taking hourly surface zonal and meridional current fields from the 1.5 km resolved North-West Shelf FOAM-AMM15 model. Since the November 2020 update, these current fields are passed to the wave model directly, as part of a two-way coupling

process between the NWS wave and physical ocean models via the OASIS3-MCT coupler (Valcke et al., 2015). The coupled surface currents are interpolated directly to the SMC grid sea-points (i.e., at the 3 km and 1.5 km cell resolutions as appropriate).

The WWIII-AMM15 model is nested one-way with lateral boundary conditions supplied from the Met Office global wave model (Saulter et al., 2016). The wave boundary data are supplied as wave spectra, which are interpolated from the global model's 25 km cell resolution onto the (predominantly) 3 km open waters grid cells found at the wave model outer boundaries. In addition to the grid resolution from which the lateral boundary conditions are sourced, the global wave model does not currently include ocean forcing, which will further serve to smooth input wave energy at the boundary edges relative to the AMM15 grid.

II.3.5 Initial conditions

For this release, the wave model does not incorporate an observation data assimilation step when deriving its initial conditions. Instead, the initial conditions are constrained over successive cycles by including a 24 hour hindcast run of the model prior to each forecast. For Met Office wave models, the role of the hindcast is to apply analysed wind and surface current data to the wave model, so that the model is forced by the best available descriptions of atmosphere and ocean. This is an effective method of preventing any drift in wave model initial conditions as the wave model is primarily responsive to the instantaneous wind forcing. The use of analysed forcing fields therefore reduces the impact of any systematic drifts in the atmospheric model. By the same token, wave model errors are generally anticipated to be dominated by errors in the wind field after approximately 24-36 hours forecast lead time; so the benefits from using assimilation to constraining initial condition errors are unlikely to hold for forecasts beyond one to two days ahead. Surface currents are taken from the analysis step in the AMM15 physical ocean model.

II.3.6 Partitioning method

Characteristics describing individual wave components that make up a given sea-state are included in the model outputs. For example, a sea may consist simply of a single wind-sea component for which all wave energy is affected by the forcing wind, or multiple swell components which have been remotely generated by distant storms. In WWIII-AMM15 these components are determined using a two-stage process (Figure 3). Individual components are derived from the two-dimensional wave spectrum following a water-shedding method as described by Hanson and Phillips (2001). This process effectively treats the wave spectrum as a topographic map from which individual peaks in wave energy can be identified in order to define the separate wave components.

The second part of the procedure follows an assumption that wind sea should be defined as only that part of the wave energy spectrum which is directly forced by the wind (this is an assumption which is most regularly used by operational wave forecasters who wish to be able to reference the evolution of wind sea directly against evolution in the local wind conditions). Using this assumption, wave spectrum bins where wave speed is slower than the (co-directed) wind speed are associated with the wind sea component. In the example in Figure 3, 15 knot easterly winds (represented by the barb symbol) force the growth of westward propagating wind-waves, whilst southerly and northerly crossing swells are also present. The wind-sea partition (W) is identified as any energy in high wave frequencies above the cut-off line, which defines where wave speeds will be slower than the (co-directed) wind speed. The remaining energy is associated with a primary swell partition (1, the northerly swell) and a secondary

swell partition (2, the less energetic southerly swell) according to the water-shedding method. The assignment of special energy to wind sea overrides any previous assignment of wave energy to the topographic components made in the first step.

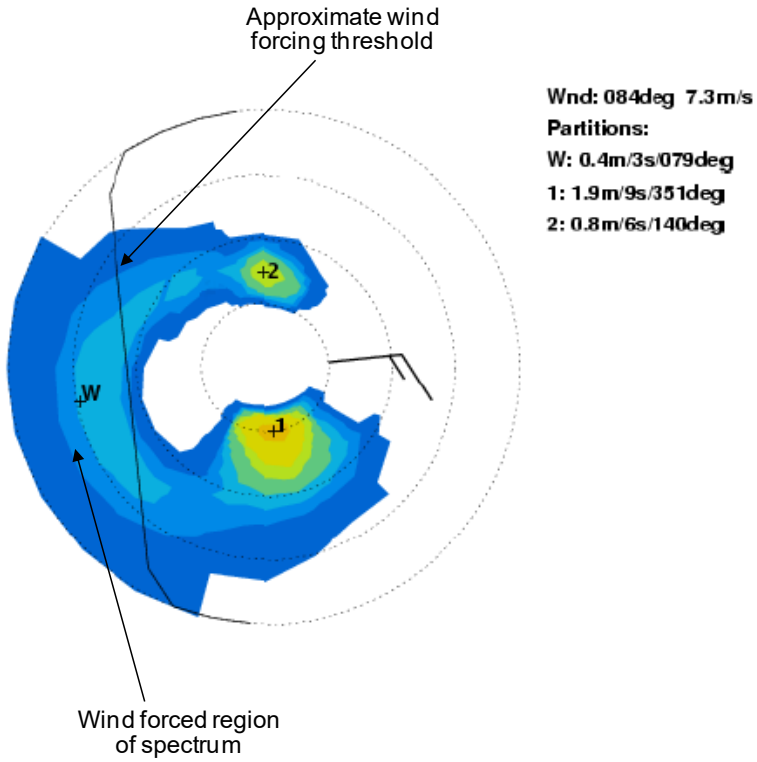

Figure 3: Schematic representation of partitioning applied to a complex wave spectrum described using energy contours on a polar (frequency-direction) plot. In the example, 15 knot easterly winds (represented by the barb symbol) force the growth of westward propagating wind-waves ('W'), whilst southward ('1') and northward ('2') crossing swells are also present.

III VALIDATION FRAMEWORK

III.1 Description of the product quality assessment

A two-year trial of WWIII-AMM15 has been carried out in order to assess the quality of the product relative to available observations. At its most basic level, the model describes the distribution in frequency-direction space of wave energy calculated over a sample of individual waves in a given sea-state. This is also known as the wave spectrum. The energy distribution is then characterised using a set of parameters (integrated from the spectrum) which can be linked to either automated or manual observations of the waves. These parameters make up the wave product set provided through the CMEMS.

Works at the Met Office to fully exploit CMEMS observation databases is ongoing. In lieu of this work being completed and quality checked, the assessment in this QuID has used better known observation sources readily available to the Met Office (see subsection III.3.1). For the North-West Shelf region, statistically robust samples of wave observations have been validated for the following parameters:

- Significant wave height (representing the mean value of highest one-third of wave heights and approximated from the zeroth moment of the wave spectrum);
- Peak period (representing the period of the most energetic waves in the sea-state and derived from the wave spectrum using the most energetic wave frequency);
- Average zero-upcrossing period (representing the time between exceedances of mean water level on the front face of successive waves and approximated using the second and zeroth moments of the wave frequency spectrum);
- Average wave direction for a subset of coastal wave buoys (representing the energy-weighted distribution of directions from which waves come and approximated using the first directional moments of the wave spectrum).

Ocean surface waves are an example of a 'forced-dissipative system' in which the wave growth and dissipation is a response to the atmospheric forcing that is applied to the ocean, whilst the waves also propagate energy away from the forcing zone. In practice, the system's primary sensitivity is to the wind forcing such that (in modelling terms) the potential for any wave forecast to drift is dominated by the behaviour of the forcing atmospheric model. On this basis, the quality of the wave model can be assessed using a hindcast generated by applying a 'best available' wind forcing dataset (in this case assumed to be the Met Office UM short-range forecast fields) and comparing resulting wave parameters with observations. Since the model is run without data assimilation, this is a true test of the wave model physics and numerical schemes under conditions assumed to be the best available representation of the true atmosphere and ocean state. By the same token, performance measures at forecast lead times effectively test the response of the wave model to any systematic drifts and errors in the forcing models. Such assessments are therefore reserved for statistics supplied within the CMEMS operational systems performance framework and are not included in this QuID.

The CMEMS NWS products are provided from the full AMM15 model domain (Figure 2). However, to reflect the complexity and variation of wave climates within the domain, performance statistics are provided for both an aggregation over the whole domain and using a number of defined shelf sea sub-regions (listed in Table 4 and visually depicted in Figure 4). Additionally, the validation sub-regions include a number of coastal areas (listed in Table 5 and visually depicted in Figure 5).

The wave model parameters that have been assessed are derived from moments of the wave spectrum. However, the method of derivation for the observed equivalent parameters may vary and is not always known from the observation metadata. As a result, this report refers to generic (rather than spectral moment specific) terms to describe each parameter, in order to identify the sea-state characteristic in

descriptive terms familiar to users. Table 3 cross-references these descriptive terms with respect to data in North-West Shelf MFC wave product files.

Table 3: Terminology used to cross-reference product quality and data names

Generic name	Model variable name	CMEMS convention long name
Signficant wave height (Hs)	VHM0	Spectral significant wave height (Hm0)
Peak wave period (Tp)	VTPK	Wave period at spectral peak/peak period (Tp)
Average zero crossing wave period (Tz)	VTM02	Spectral moments (0,2) wave period (Tm02)
Average wave direction (D_{mean})	VMDR	Mean wave direction from (Mdir)

Table 4: Extents used for regional breakdown of 'open waters' in the North West Shelf AMM15 wave domain

Region name	SW corner latitude	SW corner longitude	NE corner latitude	NE corner longitude
North-West Shelf Offshore	45.0 N	20.0 W	63.0 N	13.0 E
Northwest Approaches	53.0 N	19.0 W	61.0 N	6.0 W
Southwest Approaches	43.0 N	19.0 W	52.0 N	8.0 W
North Sea Approaches	59.0 N	5.0 W	63.0 N	4.0 E
Central North Sea	55.0 N	0.5 W	59.0 N	8.5 E
Southern North Sea	51.0 N	1.0 W	55.0 N	9.0 E
English Channel	48.5 N	6.0 W	51.0 N	1.0 E
Bay of Biscay	45.0 N	8.0 W	48.0 N	2.0 W
Celtic-Irish Seas	50.0 N	8.0 W	54.0 N	3.0 W

Table 5: Extents used for regional breakdown on UK coastal regions in the North-West Shelf AMM15 wave domain

Region name	Extent
UK Coastal	All available locations in sub-regions below
Southwest Channel	Land's End to Poole Bay
Southeast Channel	Portsmouth to Folkestone
Thames Approaches	Dover to Southwold
Wash Approaches	Happisburgh to Hornsea
Northeast England	Whitby to Moray Firth
Liverpool Bay	Rhyl to Barrow
Bristol Channel Approaches	Isles of Scilly to Ilfracombe
Bristol Channel	Minehead to Swansea

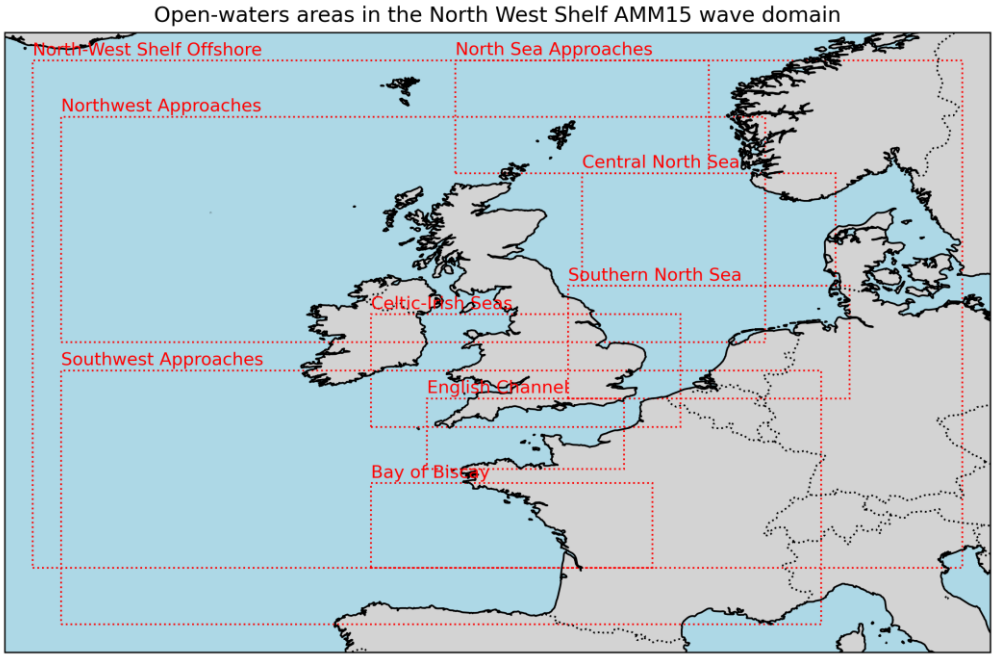

Figure 4. Extents used for regional breakdown of 'open waters' in the North West Shelf AMM15 wave domain.

Figure 5. UK coastal regions assessed in the North-West Shelf AMM15 wave domain

III.2 Description of the hindcast validation experiment

The hindcast validation experiment can be summarised as follows:

- Period of trial: 2 years, 1st January 2022 to 29th December 2023.
- Lateral model boundaries: 45°N, 20°W to 63°N, 13°E.
- Surface forcing: Met Office UM numerical weather prediction (NWP) global model forecast 10m wind fields at 10 km spatial and 1-hourly resolution; CMEMS NWS FOAM-AMM15 surface current hindcast fields at 1.5 km spatial and hourly resolution.
- Lateral boundary conditions: Wave boundary spectra from Met Office global wave model (25 km resolution at WWIII-AMM15 boundaries) run using Met Office analysis wind fields.
- Initial conditions: Model was restarted from operational restart files so spin up is not considered necessary for the WAV analysis.

III.3 Summary of tested metrics

To summarise the tested metrics, two graphical tests are applied:

1) model-observation scatter data are rendered using a 'hexbin' plot which identifies the frequency of data falling within sub-ranges of model-observation coordinate space
2) quantile-quantile (QQ) data are used to test whether the model is able to replicate the observed sample climate (effectively this relaxes the temporal and location constraints placed on data within the scatter plot).

In both cases, performance is considered good when data collapse along the 1:1 relationship between model and observation.

For all parameters, the following standard CMEMS metrics are provided:

- Mean (MEAN) of observations.
- Standard deviation (STD) of observations.
- Mean of model minus observation differences (bias).
- Standard deviation (STD) of model and observation differences.
- Root mean square difference of model and observations (RMSD).
- Pearson Correlation coefficient (CORR) between model and observations.

In addition, two further derived statistics have been provided in order to enable comparison between data from this QuID and metrics provided in support of WMO wave forecast systems (verification charts[4] for example); these include :

- Scatter Index (SI): describing a comparison between model-observation difference, standard deviation and an observed background value. For WMO wave forecast verification, the background is defined as the observed mean. In this QuID however, the observed standard deviation has been applied as the background value. The reason to adopt this convention is two-fold: i) using standard deviation compares the model-observation differences directly to the observed signal and avoids the metric being artificially deflated for parameters with a potentially high mean but low standard deviation (e.g., direction and peak period); ii) a normalisation by the observed standard deviation allows some evaluation of model skill. This is because the resulting metric is equivalent to the RMSD that would be obtained if the observed mean value was used for prediction.

[4] See: http://www.ecmwf.int/en/forecasts/charts

- Symmetric Slope (Sym. Slope): defined as the STD of the model data divided by the STD of the observations.

III.3.1 Observation datasets

The parameters and first sources of observations which were used in the quality assessment are identified in Table 6. The de-facto standard for wave measurement is in-situ data, using various instruments mounted on floating buoys and fixed marine platforms (e.g., oil installations). For open waters, a global dataset of these measurements is collated and quality controlled each month under the World Meteorological Organisation – International Oceanographic Commission (WMO-IOC) Joint Commission On Marine Meteorology's operational Wave Forecast Verification Scheme (JCOMM-WFVS; Bidlot et al., 2007). Over 400 measurement sites are registered in the system; although these are predominantly based in the northern hemisphere and within several hundred kilometres of the coasts of North America and Europe. The locations of JCOMM-WFVS in-situ data used in this product quality assessment are provided in Figure 6. Observations are sampled on a 6-hourly basis.

A substantial portion (approximately 80%) of WFVS open water observations is missing in this version of the QuID, compared to the last Met Office WAV QuID. This has limited the ability to assess peak wave period and zero crossing wave period robustly in open-waters using in-situ data. The assessment of significant wave height for open-waters is considered to be reliable with the number of available WFVS and satellite altimeter data, but peak wave period observations from WFVS number only about 9,000, which is strongly suspected to be the reason for much lower correlation achieved compared to previous versions. Zero crossing wave period observations from WFVS are completely absent currently. In UK coastal waters, a full complement of wave observations was available and verification of wave period along with other parameters is considered to be statistically robust.

The second source of observed open waters observations was taken from satellite altimeter missions. These were originally sourced from an aggregated, quality controlled and calibrated multi-mission archive developed by CERSAT (Queffeulou, 2013). During the period used for product quality assessment, data were available from the JASON-2, CryoSat and SARAL-AltiKa missions, covering the period from January 2022 until December 2023. The altimeter significant wave heights used are corrected according to a calibration provided in the CERSAT archive (Queffeulou, 2013).

Observations for the coastal waters assessment were sourced from a collection comprising Waverider buoys, in turn sourced from a number of UK coastal observatory programs, in particular the Channel Coast Observatory[5] and the CEFAS Wavenet[6] network of buoys. Locations of the coastal sites are shown in Figure 6. Observations used in these match-ups (Section III.4) are sampled on an hourly basis.

Some pre-processing and quality control of the wave observations has been applied. Experience in using the JCOMM-WFVS in-situ dataset at the Met Office has shown that significant wave height can be directly compared at the standard observation scale (representing a 20-30 minute sample of the wave field at a point). However, variations in the degree to which both these data and, more significantly, period data are discretized (period discretization varies from 0.1 to 1.0 seconds) means that more stable statistics are obtained when using the 4-hour averages described in Bidlot and Holt (2006).

Similarly, standard 1 Hz altimeter observations of significant wave height are subject to a high degree of sampling noise. At 1 Hz the altimeter 'looks' over a smaller sample of individual waves than would be measured by the in-situ sensors. Therefore, satellite altimeter data are 'super-observed' to a 3 Hz sample in order to smooth the observations and achieve a representative sampling scale closer to that of the

[5] See: http://www.channelcoast.org/
[6] See: https://www.cefas.co.uk/cefas-data-hub/wavenet/

in-situ data. In addition, prior to super-observation, two quality control measures recommended by Meteo France are used to remove small measured significant wave heights (less than 0.1 m) and measurements where the 1 Hz sample standard deviation is large (greater than approximately 1.25 m).

Less restrictive controls were applied to the coastal in-situ datasets. In particular, no time-averaging of the data was performed, due to the potential high frequency variability introduced by tidal effects. Whilst this ensures that the observed data are not aliased and that tidal signals are not removed, some degradation of the resulting averaged metrics might be expected compared to the open water sites; for example where peak periods are clearly discretised in the coastal observations dataset (see Section IV.2).

Table 6: Parameters verified and availability of supporting observations

Parameter	Coverage	Supporting Observations
Significant wave height	Full domain	In-situ platforms
Significant wave height	Full domain	Satellite altimeter
Wave peak period	Full domain (predominantly coastal, none along French coasts)	In-situ platforms
Average zero crossing wave period	Full domain	In-situ platforms
Average period of highest 1/3 of waves	Bay of Biscay	In-situ buoys
Mean wave direction	UK coastal	In-situ buoys

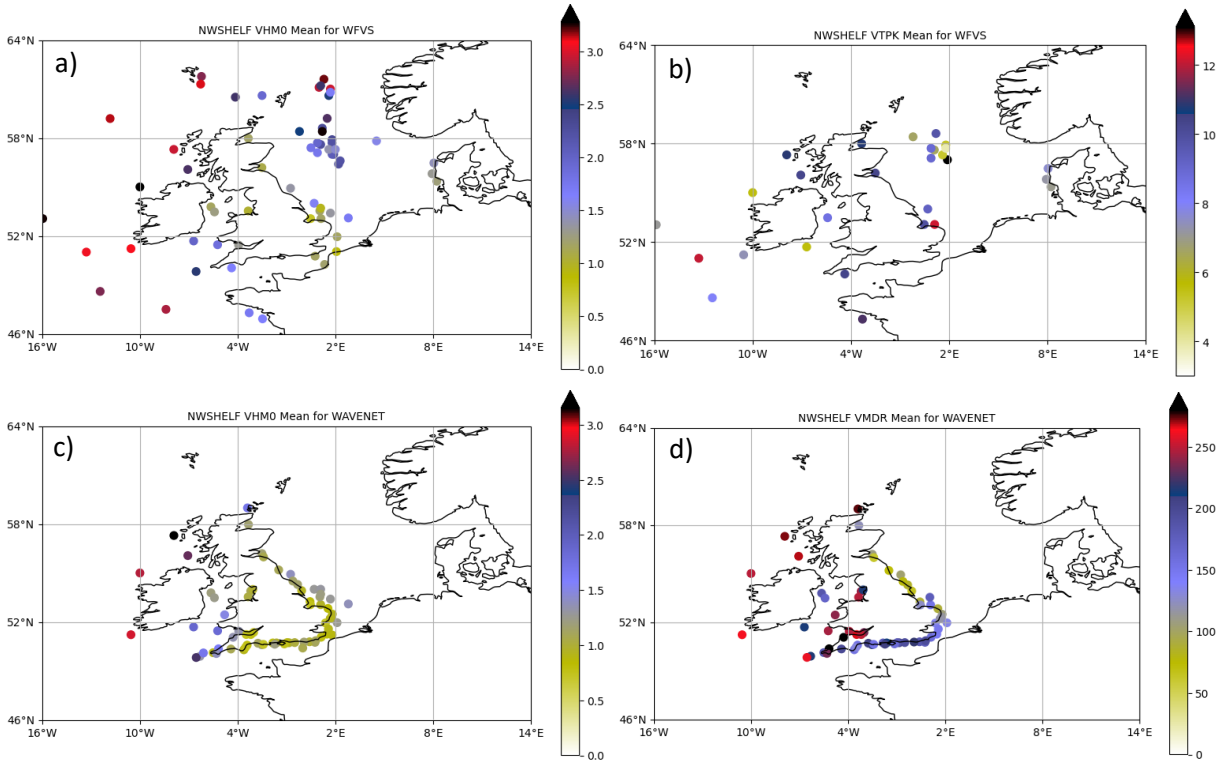

Figure 6: Mean observed parameter values at locations where in-situ data are available. a) significant wave height (VHM0) from JCOMM-WFVS collection; b) peak wave period (VTPK) from JCOMM-WFVS collection; c) significant wave height (VHM0) from UK coastal collection (all coastal parameters use these locations); d) wave mean direction (VMDR) from UK coastal collection.

III.4 Match-up and sample rejection

'Match-ups' between model predictions and real-world measurements are a key step in the validation process, requiring a routine that co-locates the model predictions in space and time with the real-world (wave buoy or satellite altimeter) measurements, thereby enabling a robust comparison of the predicted and observed values.

Match-up for all observation data has been carried out on a nearest model grid-point and time basis with a maximum distance tolerance of 10 km and, time tolerance of 1,200 seconds. In generating the final CMEMS North-West Shelf data product, the native model data are first re-gridded from the SMC multi-resolution grid to a regular grid on the model's rotated pole and, subsequently interpolated onto the final standard latitude-longitude product grid. For these experiments, match-ups were made using raw SMC grid node locations. Similar experiments run by the Met Office, using both raw SMC grid data and interpolated regular grid data indicate that validation results for the raw and re-gridded model data are nearly identical.

Analysis of coastal data, which included a substantial sample of low energy wave conditions (significant wave height less than 0.4 m), revealed a direct link between model's mismatches and low wave energy (Saulter et al., 2017). This is believed to be a result of variability in buoy response in low energy conditions; for example, where buoy motions that are genuinely due to waves may be weak compared to motions induced by other environmental factors such as wind or from the platform's own inertia. Within the validation here any match-ups with an observed significant wave height less than 0.4 m have, therefore, been rejected from the analysis.

IV VALIDATION RESULTS

This section of the document presents the results of the validation of model predictions against real-world observations. For all variables, model versus observation data scatter and quantile-quantile (QQ) relationships are shown for an overall aggregation of open waters (North-West Shelf Offshore) and coastal (UK Coastal) match-ups. In order to quantify sub-domain variability, regional statistics are provided in the accompanying tables.

Each figure in this section (see below) follows the same format. The hexbin scatter diagram illustrates the distribution of location and time matched data, with colour denoting population size. The black symbols show the (sample climate) QQ comparison, which relaxes both location and time matching criteria such that the quantiles represent the sample climate from all matched data within the specified domain. For the QQ data, each change in symbol denotes an order of magnitude increase in the percentile resolution, which is used to focus attention on the upper tail of the distribution (e.g., the change from squares to triangles defines a shift from every 1% in the data sample over the range 0-99% to every 0.1% over the range 99-99.9%). The red dashed line illustrates a linear best fit for the scatter data, where the model prediction is defined as the independent variable. Statistics describing the observed, modelled and model-observation error distributions are provided in the panel on the right of each plot.

IV.1 Significant wave height (VHM0)

Significant wave height comparisons against open waters in-situ and satellite altimeter data are shown in Figure 7 and Figure 8 respectively. For both comparisons, the QQ plots show an excellent agreement between the modelled and observed sample climate up to the 99th percentile of the upper tail (values up to 6 m). A comparison of the bias statistics indicates a small over-prediction bias (0.06 m overall) against the observations. However, reviewing the QQ data suggests that the low overall bias is as a result of compensating errors (small over-prediction for wave heights below 5 m and under-prediction above 5 m), whilst the satellite altimeter comparisons indicate a more substantial model under-prediction (0.5 m or greater) for predicted wave heights greater than 10 m (>99.8th percentile). For example, significant wave heights of up to 16 m were observed by the satellite altimeter data, while modelled significant wave height did not exceed 14 m.

A notable feature from the regional breakdowns in Table 7 and Table 8 is the increased levels of scatter in constrained shelf seas areas (Celtic Seas, Southern North Sea, and English Channel) compared to those parts of the domain in deeper waters and further from land masses (e.g., Northwest, Southwest and North Sea Approaches, see Figure 4). A number of potential sources of error may be responsible for this, including decreased skill in representing wind field and fetch details close to the land-ocean boundaries and incomplete representation of shallow water and topographic sheltering effects in the wave model. Nevertheless, the model and observed data are highly correlated (0.96 or above) in all regions.

Figure 9 and Table 9 capture a reduction in wave energy and variability observed at the UK coastal sites, resulting from shortened fetches and/or shallow water dissipation of wave energy in these waters. Bias is of a similar order to the open waters data (0.06 m overall), whilst RMSD is reduced in these regions (0.23 m overall). The QQ plot for the aggregated coastal data indicates similar bias patterns to the open waters comparison, with slight overprediction bias apparent for significant wave heights below 3 m, while above 4 m (>99.8th percentile) there is a tendency for the model to under-predict. Scatter index is increased compared to open waters, indicating that the reduction in RMSD is not proportionate to the reduction in overall variability of the wave field. This is to be expected due to the increasing complexity of processes that the model needs to represent in the near coastal zone. Correlation between the

observed and modelled data in the majority of regions is high (varying between r = 0.93 and 0.99); although is reduced in the Thames Approaches (r = 0.88) and Bristol Channel (r = 0.85). These latter areas represent tidal inlets, subject to both strong currents and where the geometry of local fetches can vary significantly, both with wind direction and through the course of a tidal cycle due to wetting and drying of the bathymetry. The poorer performance of the model in these regions indicates that the North-West Shelf wave system is not fully mature as a coastal wave model at this iteration[7], such that users should exercise caution with data in these and similar regions of the model that are subject to particularly strong tidal and wind fetch variations.

[7] The NWS-MFC are planning an upgrade to the AMM15-WWIII system to account for Wetting and Drying (Nov 2026 EiS), which will be an important step towards a coastal ocean-wave system. This is expected to improve the tidal representation and wave refraction behaviour in shallow, tidally affected model areas, and therefore may yield improvements in the wave field in coastal regions.

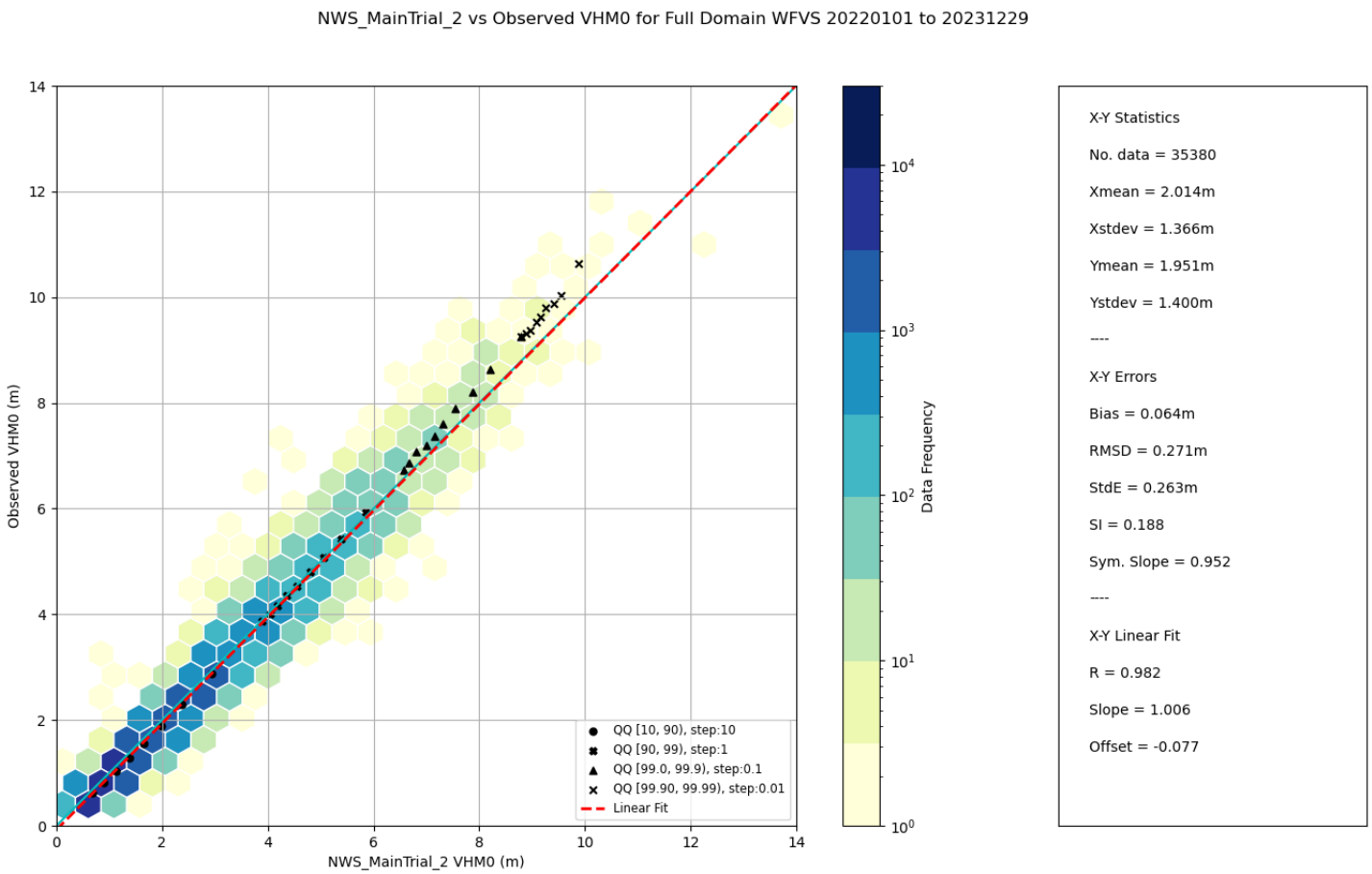

Figure 7: Scatter, quantile-quantile (QQ) and statistics results for significant wave height comparison versus open waters in-situ observations.

Table 7: Verification statistics for significant wave height comparison versus open waters in-situ observations. Units, where appropriate, are in metres. Please refer to Section II.3 for definitions of tested metrics. SI: Scatter Index; Sym. Slope: Symmetric Slope; CORR: Pearson correlation coefficient.

Region	No. Samples	Observed MEAN	Observed STD	Bias	Error RMSD	Error STD	SI	Sym. Slope	CORR
NW Shelf Offshore	19005	2.197	1.501	0.052	0.277	0.272	0.181	0.925	0.984
Celtic-Irish Seas	4551	1.417	0.903	0.050	0.242	0.236	0.262	1.152	0.971
Central North Sea	4364	1.744	1.078	0.090	0.245	0.228	0.211	0.981	0.977
Southern North Sea	4456	1.344	0.805	0.068	0.212	0.201	0.250	1.003	0.969
North Sea Approaches	2264	2.695	1.584	0.055	0.356	0.352	0.222	0.829	0.977
UK Northwest Approaches	5042	3.032	1.652	0.004	0.309	0.308	0.187	0.889	0.983
UK Southwest Approaches	3094	3.107	1.650	0.096	0.330	0.316	0.191	0.920	0.982
English Channel	1079	1.175	0.700	-0.013	0.165	0.165	0.235	1.015	0.973
Bay of Biscay	2236	2.094	1.331	0.125	0.302	0.275	0.207	0.960	0.978

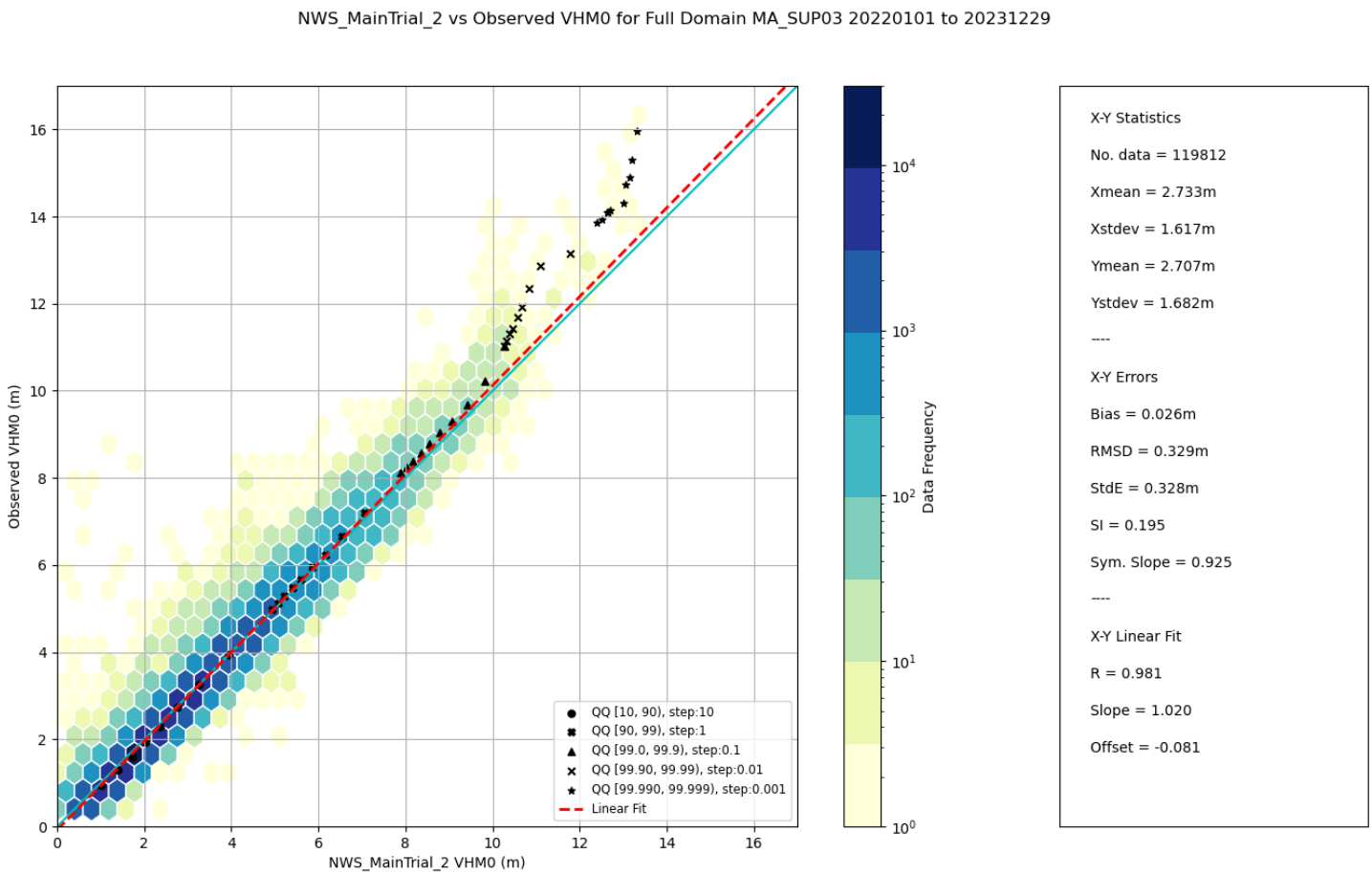

Figure 8: Scatter, quantile-quantile (QQ) and statistics results for significant wave height comparison versus open waters satellite altimeter observations.

Table 8: Verification statistics for significant wave height comparison versus open waters satellite altimeter observations. Units, where appropriate, are in metres. Please refer to Section II.3 for definitions of tested metrics. SI: Scatter Index; Sym. Slope: Symmetric Slope; CORR: Pearson correlation coefficient.

Region	No. Samples	Observed MEAN	Observed STD	Bias	Error RMSD	Error STD	SI	Sym. Slope	CORR
North-West Shelf Offshore	119812	2.707	1.682	0.026	0.329	0.328	0.195	0.925	0.981
UK Northwest Approaches	36949	3.373	1.831	0.019	0.357	0.357	0.195	0.893	0.981
UK Southwest Approaches	18105	2.987	1.649	0.084	0.354	0.344	0.208	0.912	0.978
North Sea Approaches	19388	2.783	1.540	-0.017	0.330	0.330	0.214	0.893	0.977
Central North Sea	12893	2.016	1.215	0.003	0.252	0.252	0.207	0.908	0.979
Southern North Sea	6644	1.491	0.879	0.033	0.241	0.239	0.272	0.959	0.962
English Channel	3323	1.823	1.138	0.080	0.313	0.303	0.266	1.097	0.967
Bay of Biscay	6519	2.431	1.421	0.045	0.306	0.302	0.213	0.905	0.977
Celtic-Irish Seas	3710	1.892	1.217	0.070	0.300	0.291	0.239	1.067	0.973

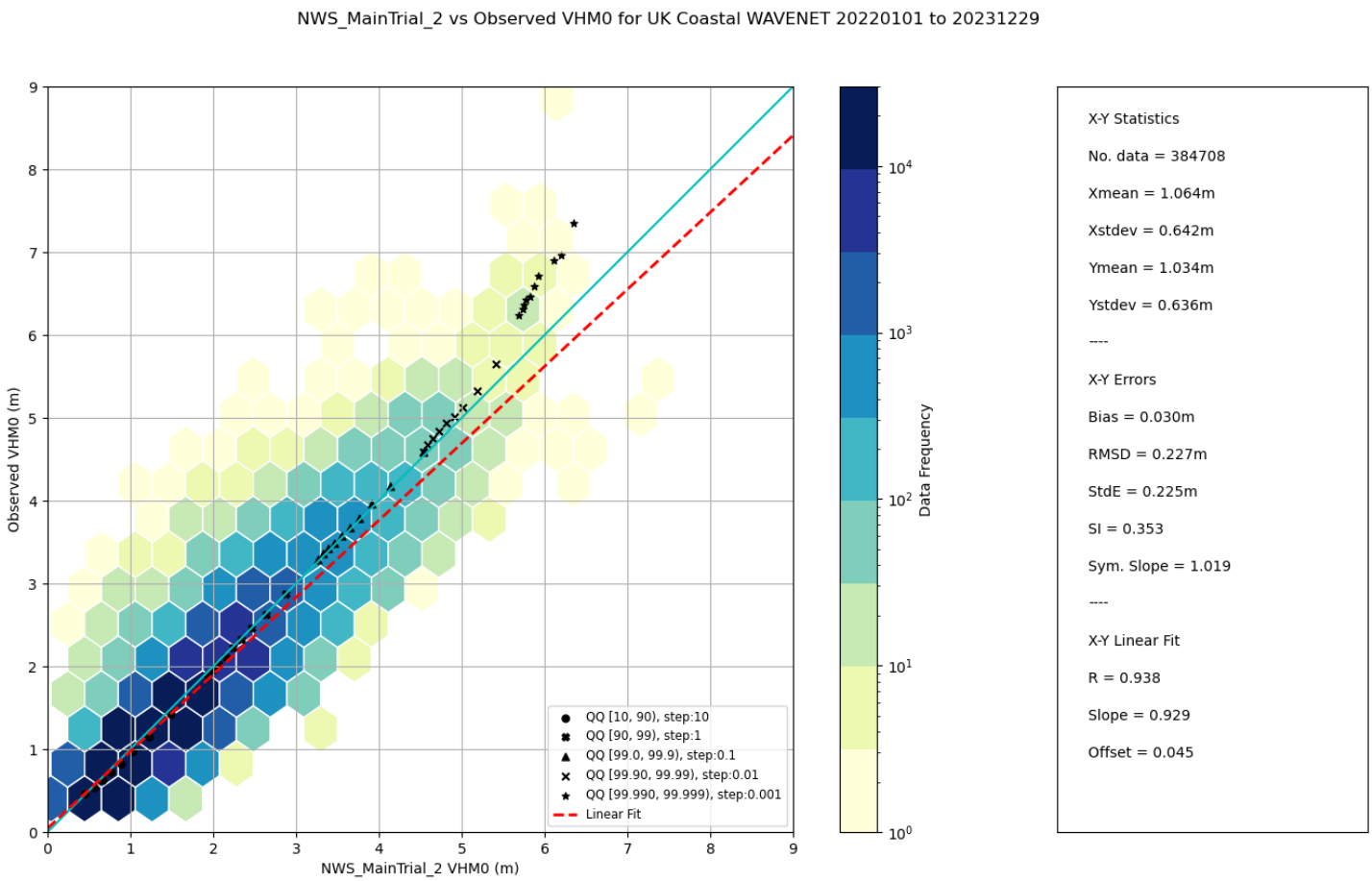

Figure 9: Scatter, quantile-quantile (QQ) and statistics results for significant wave height comparison versus UK coastal in-situ observations.

Table 9: Verification statistics for significant wave height comparison versus UK coastal in-situ observations. Units, where appropriate, are in metres. Please refer to Section II.3 for definitions of tested metrics. SI: Scatter Index; Sym. Slope: Symmetric Slope; CORR: Pearson correlation coefficient.

Region	No. Samples	Observed MEAN	Observed STD	Bias	Error RMSD	Error STD	SI	Sym. Slope	CORR
UK Coastal	384710	1.034	0.636	0.030	0.227	0.225	0.353	1.019	0.938
Southwest Channel	108124	1.066	0.673	-0.003	0.244	0.244	0.362	1.017	0.935
Southeast Channel	86793	0.973	0.568	0.058	0.229	0.222	0.391	1.149	0.931
Thames Approaches	52159	0.857	0.406	0.099	0.246	0.225	0.554	1.336	0.878
Wash Approaches	36074	0.984	0.522	0.067	0.160	0.145	0.278	0.975	0.961
Northeast England	38564	1.220	0.762	0.002	0.176	0.176	0.231	0.809	0.976
Liverpool Bay	24414	1.058	0.614	-0.039	0.321	0.318	0.187	0.839	0.985
Bristol Channel Approaches	43841	1.430	0.852	-0.027	0.183	0.181	0.295	1.068	0.958
Bristol Channel	20503	0.750	0.321	-0.005	0.240	0.240	0.282	0.953	0.960

IV.2 Peak wave period (VTPK)

Please note the caveat in Section III.3.1 regarding the reduced availability of WFVS wave period observations with which to verify modelled peak wave period in open waters areas.

Descriptions of the product quality for open waters in-situ locations are provided in Figure 10 and Table 10. Figure 10 shows significant amounts of scatter between model and in-situ observations, which is further supported by the poor correlations observed in Table 10. Whilst the scatter can be partially attributed to model error, this is significantly enhanced (compared to other parameters) due to the behaviour of the peak wave period parameter in multi-modal sea states. For example, in the case where a sea comprises a long period swell with similar energy to a developing wind-sea, the peak in wave energy identified by the model may be drawn from one component whilst the observed peak may be drawn from the other. This can lead to a large mismatch in the period even though the levels of energy assigned by the model to each component in the wave spectrum are close to those observed. These issues aside, the main population of model-observation match-ups are in good agreement, as can be seen by the clustering of coloured cells and QQ data generally falling along the 1:1 relationship line.

Statistics in Table 10 indicate substantive over-prediction biases (up to 3 seconds), which can be mostly attributed to model over-estimation of wind-sea periods (3-7 seconds in Figure 10). Scatter index and error standard deviation are notably lower for data from the Northwest Approaches and Bay of Biscay, where longer period wave energy is experienced most regularly. In these regions, the seas are often dominated by energetic swells and the observation locations appear more exposed than in the regions on the continental shelf. This may lead to a clearer signal in peak period which the model can follow more consistently.

Validation of peak wave period in UK coastal waters is influenced by discretization in the available wave observations (Figure 11) in addition to the generally wider scatter associated with this parameter. However, the comparison with coastal data indicates a much stronger correlation with observed peak wave period (r = 0.71-0.87) than the one achieved in open waters. The overall evaluation of the coastal data indicates a slight overprediction bias (0.58 seconds), but this is as a result of compensating errors, with the QQ comparison revealing a tendency for the model to bias high in the windsea period range (4-12 seconds) while underpredicting during longer period (>14 s) conditions. The regional breakdown in Table 11 shows a mixed picture of bias, symmetric slope and correlation, such that it is difficult to suggest a meaningful and physically consistent reason for errors. One clear result is a lack of model skill in representing peak wave period in the Bristol Channel (Figure 12), where the model has lower correlation (r = 0.58) and identifies significantly more events with a peak wave period greater than 5 seconds than are found in the observations.

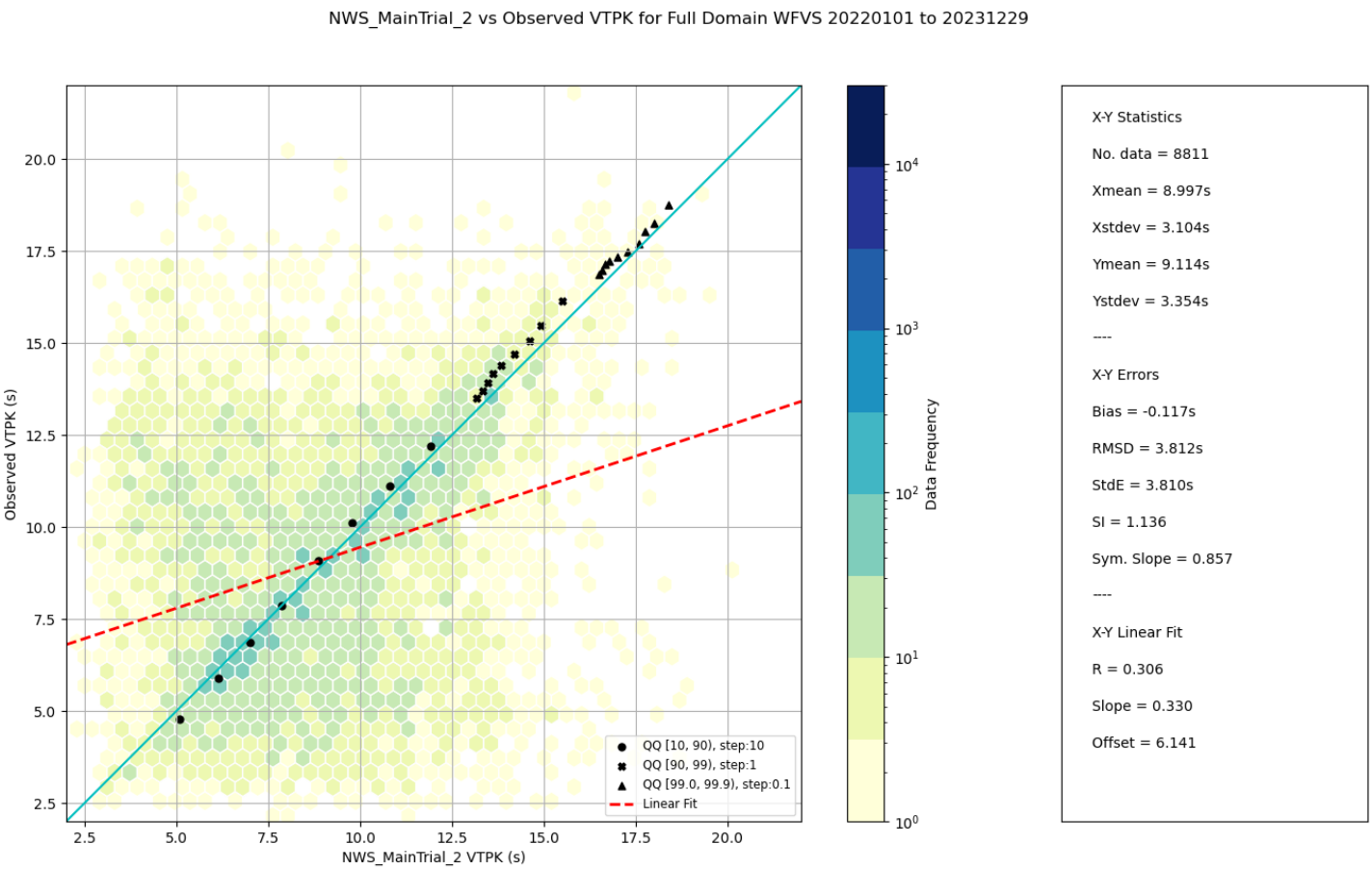

Figure 10: Scatter, quantile-quantile (QQ) and statistics results for peak wave period comparison versus open waters in-situ observations.

Table 10: Verification statistics for wave peak period comparison versus open waters in-situ observations. Units, where appropriate, are in seconds. Please refer to Section II.3 for definitions of tested metrics. SI: Scatter Index; Sym. Slope: Symmetric Slope; CORR: Pearson correlation coefficient.

Region	No. Samples	Observed MEAN	Observed STD	Bias	Error RMSD	Error STD	SI	Sym. Slope	CORR
NW Shelf Offshore	3957	8.794	3.52	1.292	3.813	3.587	1.019	0.621	0.37
Celtic-Irish Seas	288	6.175	2.038	2.861	5.105	4.227	2.075	2.454	-0.271
Central North Sea	2709	7.291	2.972	0.496	3.293	3.255	1.095	0.754	0.319
Southern North Sea	354	9.99	3.178	-1.611	4.507	4.209	1.324	0.864	0.059
UK Northwest Approaches	2127	9.829	3.238	1.232	3.084	2.828	0.873	0.513	0.524
UK Southwest Approaches	473	7.808	3.251	3.027	4.48	3.304	1.016	0.422	0.299
Bay of Biscay	1137	10.615	2.82	0.378	2.713	2.687	0.953	0.832	0.506

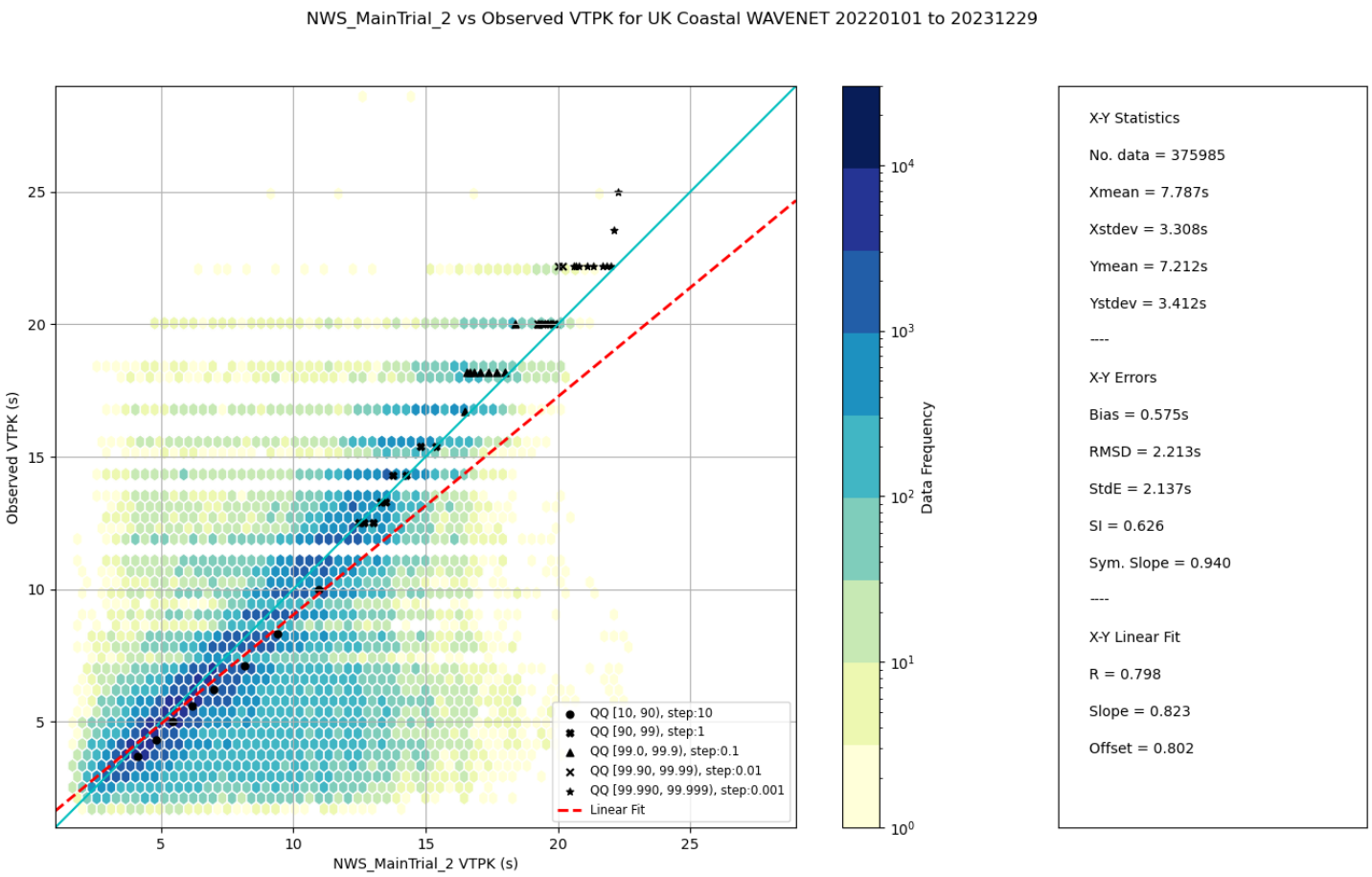

Figure 11: Scatter, quantile-quantile (QQ) and statistics results for peak wave period comparison versus UK coastal in-situ observations.

Table 11: Verification statistics for wave peak period comparison versus UK coastal in-situ observations. Units, where appropriate, are in seconds. Please refer to Section II.3 for definitions of tested metrics. SI: Scatter Index; Sym. Slope: Symmetric Slope; CORR: Pearson correlation coefficient.

Region	No. Samples	Observed MEAN	Observed STD	Bias	Error RMSD	Error STD	SI	Sym. Slope	CORR
UK Coastal	375984	7.212	3.412	0.575	2.213	2.137	0.626	0.940	0.798
Southwest Channel	104562	8.122	3.766	0.751	2.534	2.420	0.643	0.865	0.781
Southeast Channel	84398	6.565	3.257	0.841	2.608	2.469	0.758	0.998	0.712
Thames Approaches	51823	5.249	1.691	0.463	1.392	1.313	0.776	1.064	0.708
Wash Approaches	35339	7.210	2.904	0.192	1.762	1.752	0.603	0.792	0.802
Northeast England	37734	7.947	3.111	0.085	1.732	1.730	0.556	0.761	0.832
Liverpool Bay	24139	4.821	1.457	0.022	1.412	1.412	0.500	0.742	0.866
Bristol Channel Approaches	42856	9.913	3.073	-0.101	0.829	0.823	0.564	0.994	0.840
Bristol Channel	20160	5.536	2.088	0.458	1.872	1.816	0.591	0.693	0.807

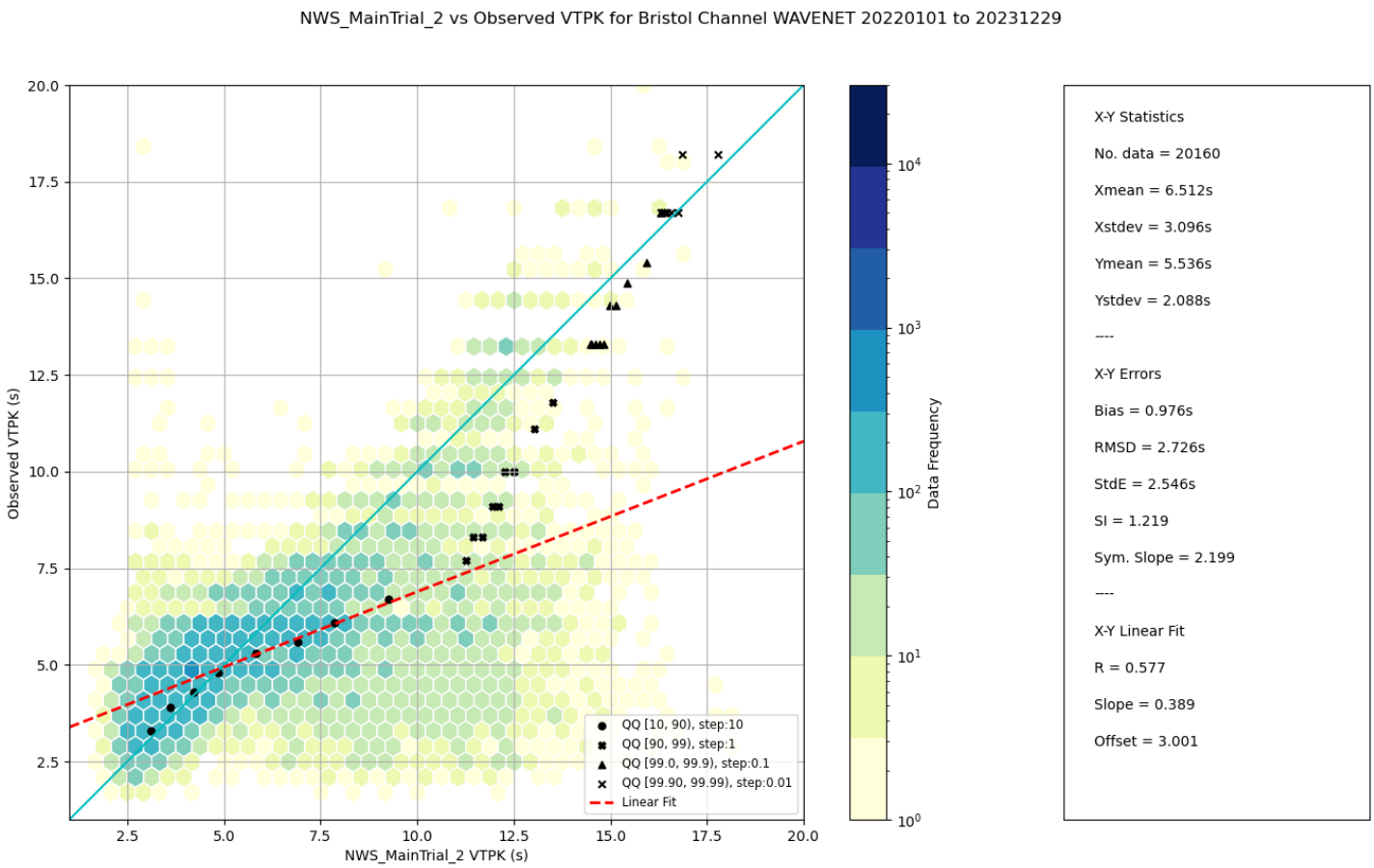

Figure 12: Scatter, quantile-quantile (QQ) and statistics results for peak wave period comparison versus UK coastal in-situ observations in the Bristol Channel.

IV.3 Average zero-upcrossing period (VTM02)

For this analysis, wave zero-upcrossing period information was only available from UK coastal buoys. Compared to peak period, the average zero-upcrossing period comparison shows more structured agreement between the model and in-situ observations at UK coastal sites, with fewer outlier comparisons (Figure 13) and slightly increased overall correlation (r = 0.57-0.91; Table 12). Overall bias is -0.06 s; however, large variation in bias is found (ranging regionally from -0.41 s to 0.10 s). The QQ comparison indicates the model generally under-predicts observations below 5 seconds and overpredicts higher wave periods by around 10-15% (Figure 13). In most UK coastal regions, while correlation is increased for zero-upcrossing period relative to the peak wave period validation, the symmetric slope for zero-upcrossing period is also higher (Table 12). This suggests an increase in the variability of the model relative to the buoy measurements.

Further review of the data reveals that this effect is specific to validation in shelf sea coasts with exposure to Atlantic swell. This is exemplified by Figure 14, which compares scatter and quantile-quantile data for the Southwest Channel and Northeast England coastal regions. Both have similar observed mean and standard deviation values but, whereas both short and long period conditions in Northeast England are dominated by locally developed seas, the Southwest Channel model data is influenced by both locally-generated windsea and swell propagating onto the shelf from deep water in the Atlantic. The most likely reasons why these conditions are not well captured by the model data are i) an under-representation of swell dissipation and ii) the challenge of correctly representing the distribution of energy with frequency during bimodal conditions in the Southwest Channel data. Across all regions and during shorter period wind-sea dominated conditions, zero-upcrossing period is slightly under-predicted (which is consistent with the validation from further offshore). These conditions are more common in the dataset than the swell events thus model biases for coastal sites are generally negative. The Bristol Channel again stands out as a region where wave period is poorly represented by the model, with the lowest correlation (r = 0.57) and highest Scatter Index (0.98) and bias (-0.41 s) of all the UK coastal regions assessed.

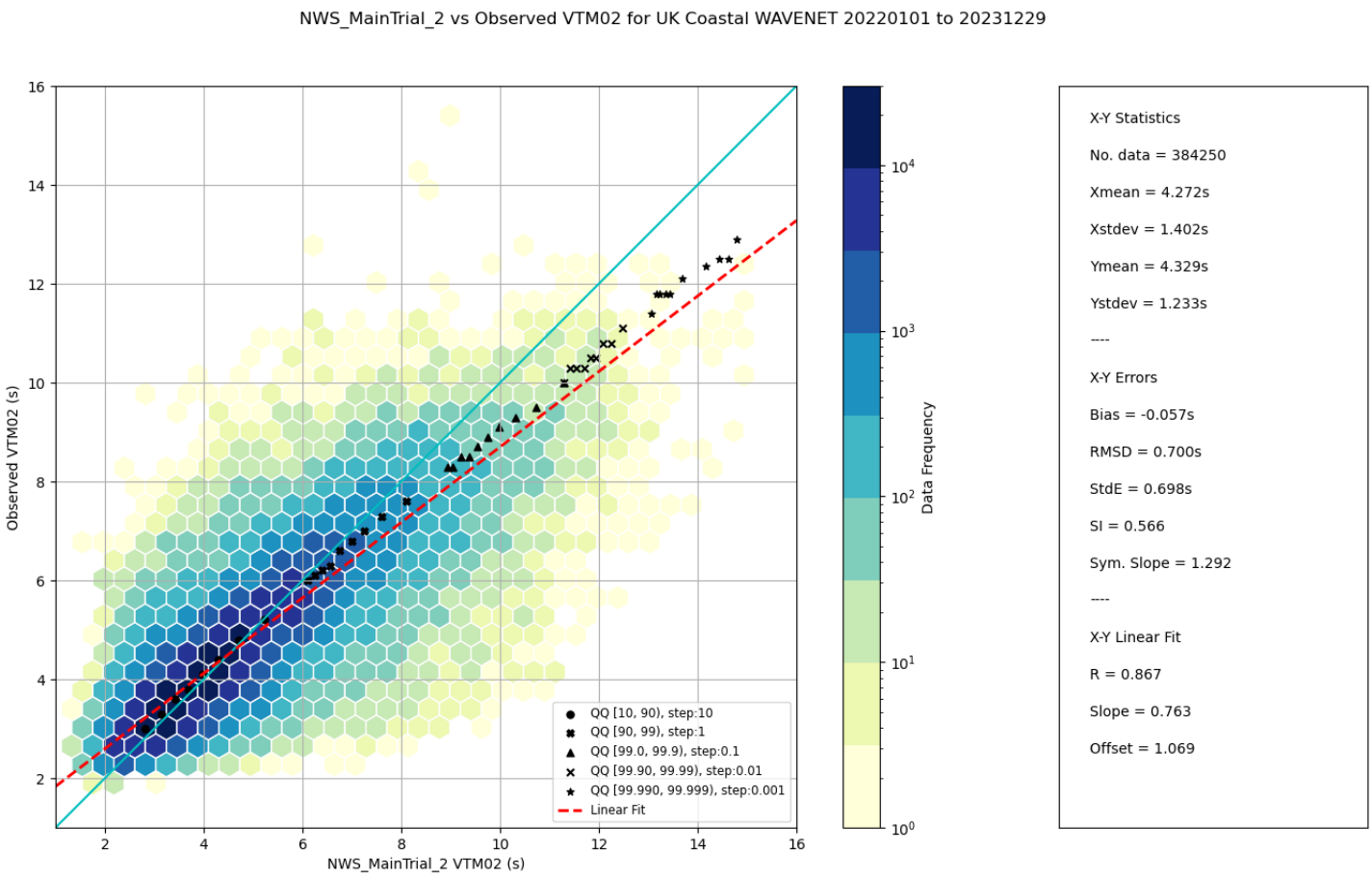

Figure 13: Scatter, quantile-quantile (QQ) and statistics results for average zero-upcrossing period comparison versus UK coastal in-situ observations.

Table 12: Verification statistics for average zero-upcrossing period comparison versus UK coastal in-situ observations. Units, where appropriate, are in seconds. Please refer to Section II.3 for definitions of tested metrics. SI: Scatter Index; Sym. Slope: Symmetric Slope; CORR: Pearson correlation coefficient.

Region	No. Samples	Observed MEAN	Observed STD	Bias	Error RMSD	Error STD	SI	Sym. Slope	CORR
UK Coastal	384249	4.329	1.233	-0.057	0.700	0.698	0.566	1.292	0.867
Southwest Channel	107949	4.54	1.225	0.057	0.769	0.767	0.626	1.413	0.850
Southeast Channel	86701	3.945	0.896	-0.039	0.628	0.626	0.699	1.411	0.809
Thames Approaches	52159	3.679	0.684	-0.096	0.537	0.529	0.773	1.051	0.709
Wash Approaches	36063	4.336	1.013	-0.126	0.547	0.532	0.525	1.145	0.873
Northeast England	38563	4.845	1.297	-0.183	0.571	0.541	0.417	0.970	0.912
Liverpool Bay	24412	3.661	0.794	-0.373	0.506	0.342	0.431	1.003	0.907
Bristol Channel Approaches	43659	5.544	1.515	0.104	0.890	0.884	0.583	1.195	0.848
Bristol Channel	20483	3.76	0.789	-0.412	0.873	0.770	0.976	1.219	0.574

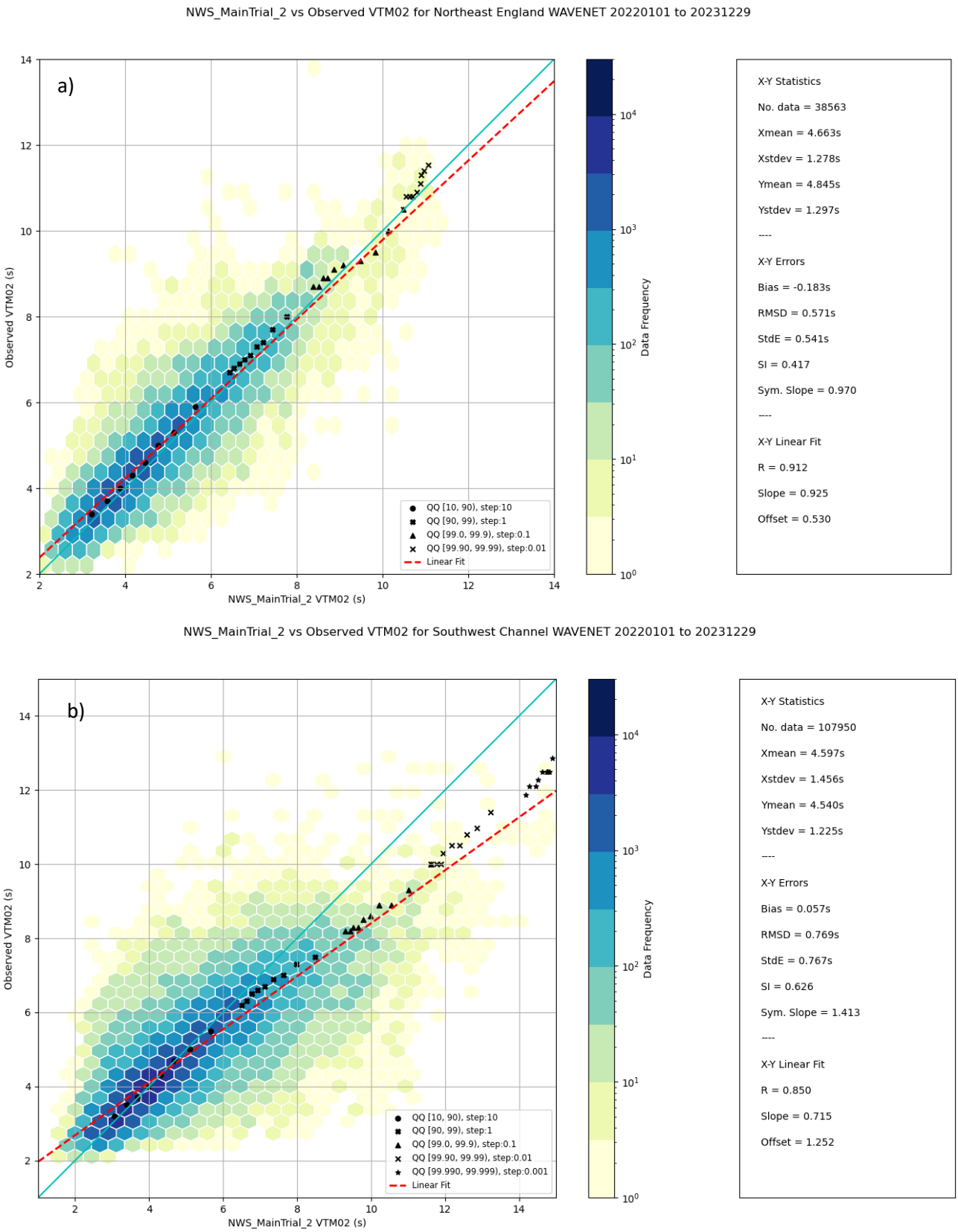

Figure 14: Comparison of scatter, quantile-quantile (QQ) and statistics results for average wave zero-upcrossing period comparison versus UK coastal in-situ observations for (a) Northeast England and (b) Southwest Channel regions.

IV.4 Average wave direction (VMDR)

For this analysis, wave direction information was only available from UK coastal buoys. Since direction is a 'circular variable', observed standard deviation, bias and RMSD are calculated using a reduction that ensures direction differences can be no larger than 180 degrees. Correlation is not supplied as a metric since values are artificially lowered when comparing model-observation match-up values falling either side of the 360 to 0 degree crossing point.

Figure 15 indicates high levels of scatter in the direction match-ups, but the majority of data (indicated by the coloured bins) are reasonably well predicted, leading to scatter index values in the range 0.28 to 0.53 (Table 13). RMSD is within the range 16-33 degrees. Directional variability in Wash and Thames Approaches (see Table 4 for a detail on the regions) is overestimated by the model possibly reflecting an under-estimation of the directional constraint introduced by refraction over shallow water bathymetry in these regions. In the Bristol Channel, the largest degree of directional bias occurs, possibly due to strong tidally induced refraction in this area being underpredicted. However, bias for the direction data in all regions is small and below the 10-degree resolution of the model spectra.

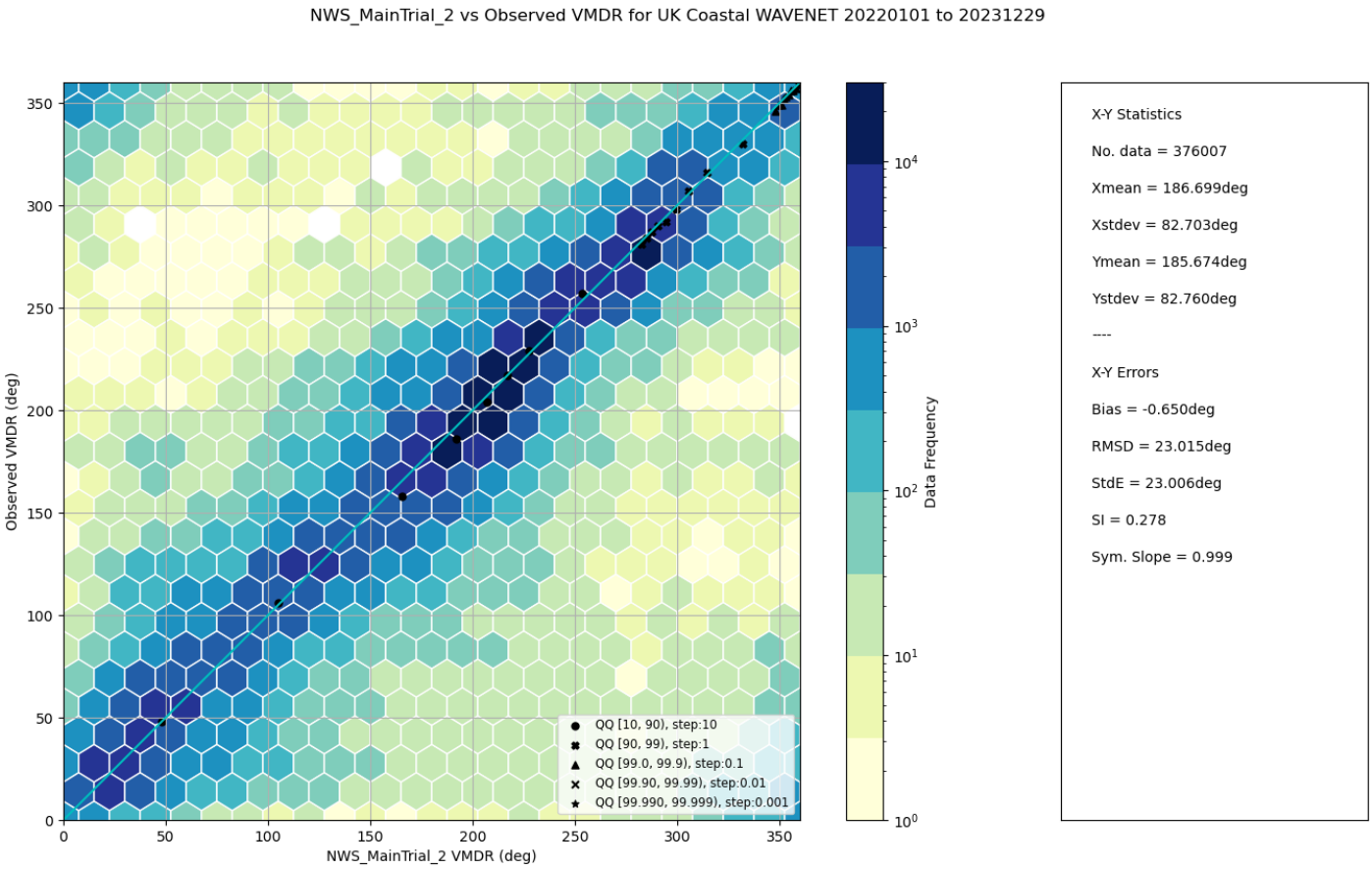

Figure 15: Scatter, quantile-quantile (QQ) and statistics results for average wave direction comparison versus UK coastal in-situ observations.

Table 13: Verification statistics for average wave direction comparison versus UK coastal in-situ observations. Units, where appropriate, are in degrees. Please refer to Section II.3 for definitions of tested metrics. SI: Scatter Index; Sym. Slope: Symmetric Slope; CORR: Pearson correlation coefficient.

Region	No. Samples	Observed MEAN	Observed STD	Bias	Error RMSD	Error STD	SI	Sym. Slope
UK Coastal	376007	185.674	82.760	-0.650	23.015	23.006	0.278	0.999
Southwest Channel	104481	199.065	37.088	-0.746	16.513	16.496	0.445	0.887
Southeast Channel	84161	186.452	50.238	3.468	21.963	21.688	0.432	1.043
Thames Approaches	51873	133.949	69.378	-5.109	32.544	32.141	0.463	1.145
Wash Approaches	35367	109.314	69.967	-0.125	30.007	30.007	0.429	1.096
Northeast England	37752	86.123	60.513	-1.130	28.316	28.294	0.468	0.905
Liverpool Bay	24298	245.080	63.372	-4.605	22.397	21.918	0.346	1.043
Bristol Channel Approaches	42771	269.286	30.996	0.912	16.426	16.401	0.530	0.932
Bristol Channel	20160	251.419	56.575	-5.432	24.123	23.504	0.415	0.888

IV.5 Conclusions of wave model hindcast performance

The base model for the NWS wave products, WAVEWATCH III, has a well-established provenance; for example being used for operational forecasts at a number of centres including the Met Office, NCEP, SHOM, Puertos del Estado and Meteo France. The source term physics package used in WWIII-AMM15 has been successfully deployed in Met Office and NCEP global wave model configurations. Compared to the default settings described in Tolman (2014) and the Met Office global model described by Saulter et al. (2016), the physics scheme has been slightly retuned in the NWS application in order to better represent short fetch wave growth commonly occurring in the shelf seas of this region.

The quality assessment of significant wave height tests the wave model's ability to represent the bulk energy imparted to ocean surface waves from the atmosphere. Results indicate that the model reflects the time evolution of the wave energy well (correlation coefficients >0.96) and with minimal systematic biases in open waters (on the order of 0.1 m). Closer to the coast, this performance is slightly reduced (correlation coefficients ranging r = 0.93-0.98). In addition, two outlier regions (the Bristol Channel and Thames approaches) are identified as having much poorer performance (r = 0.85-0.87). These are particularly difficult areas to model, due to strong tidal flows and limited fetches on certain wind directions, which are further varied due to geometry and depth changes dependent on phase of the tide.

The quality assessment of average zero-upcrossing period tests the wave model's ability to represent the overall distribution of wave energy through the frequency domain of the wave spectrum. A quality assessment of average zero-upcrossing period could only be carried out for UK coastal locations, due to a lack of available observations further offshore. The time evolution of these parameters is captured reasonably well for more exposed coasts (correlation coefficients of approximately r = 0.81-0.91). The degree of correlation is reduced in less exposed coastal regions (r = 0.57 and 0.71 in the Bristol Channel and Thames approaches, respectively). The model generally under-predicts observations below 5 seconds and overpredicts higher wave periods by around 10-15%. In coastal regions partly exposed to Atlantic swell, some reduction of performance is observed; likely due to a difference in the way the model and observations capture the balance between wind-sea and swell components of certain sea-states.

The quality assessment of peak wave period tests the model's ability to represent wave growth and dissipation processes in the frequency domain, since the peak frequency will shift from higher to lower frequencies as waves grow and the sea-state matures. In some cases, a sea-state will comprise both wind-sea and swell, in which case small mismatches in the balance of energy between these components may lead to large peak period error. For example, where the model identifies long period swell as the primary component in a sea, the observations may identify a much shorter period wind-sea as the most energetic. In the assessment here, these instances, which make a large contribution to RMSD, can be easily identified in the scatter diagrams (Figure 10 and Figure 11). In open waters, quantiles throughout the peak wave period range were well represented by the model, albeit with poor model-observation correlation (in range r = 0.00-0.52) and a relatively high bias (up to 3 s) in some regions. In coastal waters, correlation is much higher (in range r = 0.70-0.87) apart from an outlier region, the Bristol Channel, having poorer performance (r = 0.58). The model tends to bias high in the windsea period range (4-12 seconds) while underpredicting during longer period conditions. Overall, however, the results suggest that during energetic events where a sea is dominated by one particular wave component, the peak in the wave spectrum is reasonably well predicted by the model. In cases of multi-modal seas where no single component is dominant, attention should be paid to the potential effects of model uncertainty in each component identified by the model's partitioning scheme. Both these results and those for mean zero-upcrossing period indicate that there is scope to improve the shape of the underlying wave spectra generated in the model.

A quality assessment of mean wave direction could only be carried out for UK coastal locations, due to a lack of available observations further offshore. Some differences might be expected in model quality in the two regions since, whilst the coastal zone is harder to model due to wave refraction and dissipation effects in shallow water, mean wave direction offshore will be more likely to be influenced by multiple components of a sea-state; which could come from very different directions. The evaluation of model-observation scatter indicates that, in the coastal zone, there are a number of events where modelled and observed mean wave direction can be substantially different. However, RMSD and scatter index statistics are sufficiently low to suggest that such events represent noise in the overall direction time-series rather than a systematic error in model representation of this parameter. Such noise can be introduced where either rapid changes in dominant wave direction occur and the model, and observed timings of these changes are slightly different, or where the model and observed balance of sea and swell components in multi-modal sea-states introduces a sensitivity (e.g. Saulter et al., 2017).

The wave model products provided by the North-West Shelf MFC will include a number of parameters that could not be directly assessed against readily available measurements. Based on the results presented however, it seems reasonable to conclude that in the majority of cases, the accuracy values quoted for significant wave heights and period will be applicable to partitions of the wave spectrum. Regarding directions in open waters, no gross errors are expected in the wave model since wave energy development in constrained shelf seas can be highly sensitive to the fetch over which waves are developed, and this has a strong directional component. This implies that if the significant wave height performance of the model is good, then the direction information should also be correct. Quality assessment of the wave model Stokes Drift components and newly introduced maximum wave height estimates is presently considered as a subject for future investigation.

It should be noted that the 1.5 km coastal resolution of this model restricts the applicability of information provided in this document to open waters approximately 5 km or more from the coastline and where bathymetric variability is on the scale of a kilometre or greater.

V SYSTEM'S NOTICEABLE EVENTS, OUTAGES OR CHANGES

This section is dedicated to tracking and describing changes that have occurred in the NWS-WAV NRT operational systems (including model system version upgrades or changes related to upstream data updates) (Table 14).

Table 14: Noticeable changes in the product operational systems.

Date	Change/Event description	System version	Other
Oct 2018	Wave model and 1.5 km wave product introduced in North-West European Shelf MFC production system. This is a new product replacing the WWIII-AMM7 based (7 km) wave product in system WWIII-AMM7v1	WWIII-AMM15v1	
Dec 2020	Direct two-way coupling between wave and physical ocean models. Surface current fields interpolated direct to sea-points.	WWIII-AMM15v1.1	
Sep 2023	Complete system update implying change of the modelling system configuration and resolution.	MFWAM-v6 (Copernicus-2 Phase-I)	
Nov 2024	New product, at 1/36° resolution.	MFWAM-v7 (Copernicus-2 Phase-I)	
Nov 2025	Re-entry into service of Met Office coupled wave-physics analysis forecast system, including a change to Met Office UM wind forcing.	WWIII-AMM15v1.2	

VI QUALITY CHANGES SINCE PREVIOUS VERSION

This section aims to provide a short summary of the main changes in Product Quality between the previous and current version.

The present NWS-MFC system is produced by a new NWS-MFC consortium, and so it cannot be compared to the previous NWS-IBI system delivered from 2023-2025. It is a similar system to what was available prior to November 2023, which was also delivered by the Met Office. A Copernicus funded internal report compared the Met Office and NWS-IBI system and found that the Met Office system better represented the tides and SST; with the limited NWS observations however, they were otherwise similar. The new NWS-MFC system is the same system as the Met Office system but the lateral and surface boundary conditions are now provided by Met Office atmospheric and ocean systems.

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